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Correlated-Equilibrium-Based Unmanned Aerial Vehicle Conflict Resolution

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Collision avoidance is an essential component of an autonomous robotic system. This paper presents CONCORD, a novel decision-making system for unmanned aerial vehicle (UAV) conflict resolution that uses the notion of correlated equilibrium. Correlated equilibrium is a solution concept in game theory, which is computationally light and gives a better-expected payoff to every player. This paper proposes an intelligent semi-autonomous conflict resolution system for aerial vehicles. The vehicles are the players, and the resolutions are the equilibrium strategies of the game. CONCORD monitors the vehicles to predict conflicts and compute strategies leading to equilibrium. The cases of cooperative, non-cooperative, and multiple vehicles are discussed, emphasizing decision making and its corresponding timelines. Monte Carlo simulations are also performed to demonstrate the effectiveness of the proposed framework. The proposed framework is compared with two other baselines and a Nash-equilibrium-based resolution to highlight its effectiveness. This work also presents the integration of CONCORD to an existing UAV traffic management system. The proposed module would enable conflict-free flight for a multi-UAV system in an efficient manner.

I. Introduction

UNMANNED aerial vehicles (UAVs) are in demand for a wide range of applications. A traffic management system similar to the Air Traffic Control (ATC) is necessary to handle these vehicles efficiently. The complexity associated with such a system will increase with the size of airspace and the number of vehicles using it at any given instant of time. The need for collision avoidance among UAVs is quite significant for large-scale integration into the airspace. When multiple individuals are involved in decision making, the solution is quite often coupled; that is, the outcome for each individual depends upon others' decisions. Game theory is a tool to analyze such rational decision making and also predict behavior of the individuals involved. This mathematical tool models the problem as games by translating the decisions and outcomes to equivalent values. The individuals involved are players who interact with egoistic goals. Due to its ability to handle problems involving multiple rational agents, game theory finds application in a broad spectrum of fields.

Nash equilibrium is defined based on several assumptions [1], such as the agents being rational, no binding contracts, and no external intervention. There could be cases in which these assumptions are not valid. There are many methodologies that relax the existing assumptions of Nash equilibria, few of which are the Bayesian games [2] and the correlated equilibria. In this paper, we look into correlated equilibrium and its application to UAVs. Correlated equilibrium was introduced by Aumann [3] and has given rise to a substantial body of work in the literature exploring the various possibilities of this concept like existence [4], mechanisms for communication [5], learning procedures [6], and computational methods [7]. The concept of correlated equilibrium finds application in diverse domains. Optimal spectrum sharing using correlated equilibrium is discussed in [8]. An energy-efficient resource allocation game is formulated in [9], whereas a cooperative policy selection game is formulated in [10], which uses correlated equilibrium to achieve energy efficiency in ad hoc networks. A reinforcement learning-based method for power grids using correlated equilibrium is studied in [11].

Collision avoidance and traffic management of unmanned systems are of great significance as the UAVs are finding a wide range of applications [12–14]. Collision detection and resolution is a deeply researched area in robotics for both ground and aerial vehicles. A variety of algorithms exist addressing the problem that could be broadly classified into geometric methods, potential field-based methods, optimization methods, and others. Collision cones [15] and velocity obstacles [16] are algorithms belonging to the first, whereas [17,18] propose algorithms using the second category of avoidance methods. Optimization-based avoidance method is adopted in [19]. There are other methods like avoidance map [20], which are derivatives of the methods in earlier classes. Partially Observable Markov Decision Process (POMDP)-based methods [21] are increasingly being used for collision avoidance of aircraft. Any of these methods could be used as the avoidance algorithm for the proposed CONCORD framework. The Airborne Collision Avoidance System (ACAS) sXu is a detect-and-avoid framework for small unmanned vehicles. References [22–24] highlight the deficiencies of TCAS and introduce modeling of conflict as a Markov decision process. Further, the resolutions are computed using dynamic programming. Extensions to these are presented in [25,26], where a formal framework is defined for detect and avoid.

One of the early attempts to formulate the avoidance problem using game theory is reported in [27], where safe navigation of ships is discussed. Avoidance for ground vehicles is addressed using differential games in [28–30]. Nash-equilibrium-based solutions for road traffic scenarios are proposed in [31–33]. Avoidance problems of aerial vehicles are dealt with using satisficing game theory in [34], where resolutions are optimized considering all agents. Simultaneous game-theory-based UAV avoidance is presented in [35], where game trees are used to solve conflicts. A few papers model pilot behavior [36,37] using reinforcement learning and game theory to analyze the avoidance problem. Most of the reported formulations for conflict resolution achieve optimal solutions at the cost of computation. Also,

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the assumptions used are difficult to implement in realistic scenarios. Several traffic management architectures like UTM [38] and U-space [39] are already under development. UAV integration into airspace using game theory is addressed in [40,41]. The International Civil Aviation Organization (ICAO) is presently engaged in laying out foundations for rules and regulations to be followed for safe and efficient UAV traffic management [42]. As the U.S. Department of Defense (DoD) report [43] suggests, a semi-autonomous system is a logical step before stepping up to the stage of complete autonomy of UAVs.

In this work, we look into a novel semi-autonomous conflict resolution architecture that uses the concept of correlated equilibrium for UAV avoidance and safety. Our initial contribution in this direction, CONCORD (a UAV conflict resolution system using correlated equilibrium-based decision making), is reported in [44]. This paper extends the work further, dealing with multiple UAVs and defines CONCORD as a complete architecture. Multi-UAV systems are the definite future, and their full autonomy is not straightforward when vehicles have conflicting interests. The importance of mediated decision making in such scenarios and how correlated equilibrium could be employed for decision making are presented in detail with examples. The paper describes how CONCORD system ensures safety and achieves resolutions in a fair manner. The work also describes how CONCORD system could be integrated with an existing UAV traffic management system to address conflicts. Strategic deconfliction is of much importance in the current UAV traffic management scenario. The CONCORD system does not explicitly define a fairness metric. Instead, fairness is incorporated by considering UAV priorities to compute correlated equilibrium strategies. However, Evans et al. [45] define a fairness metric for resource allocation in the context of UAV traffic management.

The studies cited are those predominantly from correlated equilibrium and collision avoidance of UAVs. However, in this work we present a high-level decision-making framework that coordinates among the UAVs using any available avoidance algorithms for deconfliction. The avoidance strategies computed by each of the algorithm can be translated into a payoff function and the CONCORD algorithm works within the framework of these payoff functions. There is no similar framework in literature for fair deconfliction. Available ones are based on different, and often disparate, sets of parameters, which makes comparison difficult. A surveillance and tracking module to estimate UAV states and predict conflict and a threat resolution module to resolve conflict are introduced in [25]. The resolutions are computed using the techniques described in [24,25]. A discussion on handling uncertainties is given in [26] along with definitions of "well-clear" for UAS. In contrast, the proposed framework models the problem using game theory. The correlated strategies are resolved using linear programming. Note that the overall framework [25] has some similarity to the proposed CONCORD system in that both of these have a surveillance/monitoring unit that estimates conflicts and switches control to threat resolution module upon conflict.

The paper is organized as follows: Section II gives an outline of the concept of correlated equilibrium and collision avoidance. Section III gives the details of the underlying methodology of applying correlated equilibrium for conflict resolution and the formulation. Integration of CONCORD to an available UAV traffic management system is discussed in Sec. IV. Section V gives the numerical simulations and demonstrative examples along with Monte Carlo simulations and comparison with other methods. Section VI concludes the paper.

II. Preliminaries

This section presents the basic concepts used in the development of CONCORD, covering the formal definition of correlated equilibrium, its computation, and the idea of collision avoidance among UAVs.

A. Correlated Equilibrium

1. Definition

Correlated equilibrium [3] is a concept in game theory that generalizes the notion of Nash equilibrium. In many problems, it is found that Nash equilibria provide solutions that are not optimal compared to certain payoffs, which are nonequilibria outcomes. This suboptimal performance of Nash equilibrium has led to the notion of correlated equilibria where a mediator suggests the best actions for the players. The mediator could be a negotiator or a correlation device that could recommend the game's best options to the players. Best options are those that optimize the priorities or utilities of the players. The best response to a game gives the players the most favorable outcome. Thus, introducing correlation to the game can result in better payoffs for the players. The formal definition for correlated equilibrium is given below.

Consider an $N \geq 2$ person constant-sum game G. G is specified by strategy set $S = \{S_1, S_2, \cdots, S_N\}$, where $\{S_1, S_2, \cdots, S_N\}$ denote the action sets of each of the N players. An associated payoff function is given by $U(s_1, s_2, \cdots, s_N)$: $S_p \to R$, where $(s_1, s_2, \cdots, s_N) \in S_p$ and $S_p = S_1 \times S_2 \times \cdots \times S_N$ is the joint strategy profile set. Consider a finite probability distribution $P = (p_1, p_2, \cdots)$, which is a tuple of size equal to number of elements in the joint strategy space. Let player i be recommended to play strategy s_i with a posterior p_i and the outcome of this recommendation be $U_i(s_i, s_{-i})$, where s_{-i} is the strategy followed by those players other than player i. A correlated equilibrium of G is a correlated strategy that satisfies the following:

$$\sum_S p_i U_i(s_i, s_{-i}) \geq \sum_S p_i U_i(s_j, s_{-i}), \quad \forall\ s_j, s_i, s_{-i} \in S \tag{1}$$

These constraints are also called strategic incentive constraints [46]. The mediator recommends the correlated strategies to the players. For the players to follow the suggestion, there should be no incentive to deviate from the provided suggestion, as given in Eq. (1). Therefore, if the players are following the recommended strategies, the resulting game is said to be in correlated equilibrium.

The computation of correlated equilibrium can be demonstrated by the following example. Consider a two-player non-zero-sum game whose payoff matrix is given in Table 1. Player A chooses rows D or C while player B chooses columns D or C. The numbers in the payoff matrix (a_i, a_j) represent the measure of gain or loss that each of the players incurs while choosing a particular action pair. Both the players are utility maximizers. For the entries satisfying $0 \leq a_1 < a_2 < a_3 < a_4$, the pure strategy Nash equilibria are (D, C) and (C, D), whereas for mixed-strategy Nash equilibrium, player A mixes his/her action with probability $(q, 1-q)$, where q is given by

$$q = \frac{a_4 - a_3}{a_2 + a_4 - a_3 + a_1} \tag{2}$$

Now, assume that there is some third party or a semi-automated system that chooses a strategy pair with some prior p. Each player receives the information about his/her strategy from this external entity and implements it. Given the recommendation, there exists a conditional probability associated with other players' strategies. The strategy recommended by the third party would be optimal given this conditional probability. Table 2 gives a representative probability distribution for this two-player game. For this distribution to be in correlated equilibrium, the following equations should hold:

Table 1 Payoff matrix of two-person game

		Player B	
		D	C
Player A	D	a_1, a_1	a_4, a_2
	C	a_2, a_4	a_3, a_3

Table 2 Probability distribution for the strategies

		Player B	
		D	C
Player A	D	p_1	p_2
	C	p_3	p_4

$$
\begin{aligned}
(a_1 - a_2)p_1 + (a_4 - a_3)p_2 &\geq 0\\
(a_2 - a_1)p_3 + (a_3 - a_4)p_4 &\geq 0\\
(a_1 - a_2)p_1 + (a_4 - a_3)p_3 &\geq 0\\
(a_2 - a_1)p_2 + (a_3 - a_4)p_4 &\geq 0
\end{aligned}
\tag{3}
$$

The expected utility associated with each of the players is

$$
\begin{aligned}
\mathrm{E}U(A) &= p_1 a_1 + p_2 a_4 + p_3 a_2 + p_4 a_3\\
\mathrm{E}U(B) &= p_1 a_1 + p_2 a_2 + p_3 a_4 + p_4 a_3
\end{aligned}
\tag{4}
$$

The expected payoff associated with the correlated equilibrium, Eq. (4), would be at least as good as the Nash equilibrium payoff, as computed using Eq. (2). Introducing such correlated randomness into the game could result in better outcome for all the players. Equations (3) are used to compute the candidate probabilities, given the payoff matrix. Those optimizing some objectives can be chosen from this set of probabilities, and the resultant strategies will lead to correlated equilibrium.

While finding a Nash equilibrium solution is Non-deterministic Polynomial-time hard (NP-hard) [47], correlated equilibrium only requires solving linear inequalities. The relation between the set of correlated equilibria and Nash equilibria in such general cases is represented by Fig. 1. In other words, the set of correlated equilibrium strategies is a superset of the set of Nash equilibrium strategies. It is known that every game has at least one mixed-strategy Nash equilibrium. Thus, every game has at least one correlated equilibrium also, which is the mixed strategy Nash equilibrium. The only difference is that the mixing is independent in Nash equilibrium. This means that the players randomize in private and follow a Nash equilibrium strategy profile, mutually agreed before the game. However, in correlated equilibrium, the players could use a common source of randomization or a mediator could observe an event to recommend strategies that result in better-expected payoff for both the players. Such randomization of actions results in nonindependent mixing of strategies. Also, the fairness in distribution is achieved by choosing objective functions like the social welfare function to compute the correlated strategies. Nash equilibrium strategies may not have this fairness property. This is discussed in detail in the following sections. In essence, the major features of correlated equilibrium are as follows:

i) Correlated strategies and equilibria are more general than Nash equilibria. In Nash equilibrium, the players are expected to be rational, and the equilibrium is self-enforcing. This behavior is often unrealistic. In correlated equilibrium, a third party performs randomization and recommends strategies to players.

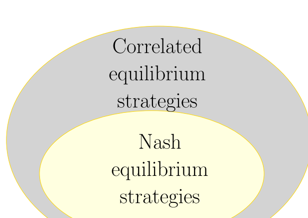


Fig. 1 Set of correlated equilibrium strategies is a superset to that of the Nash equilibrium strategies.

ii) The correlated equilibria are found by solving a set of linear inequalities, thus being computationally efficient. Nash equilibrium for multiplayer games is nonlinear and tedious to compute [48].

iii) The expected utility from correlation could result in a better payoff than that obtained from the Nash equilibrium of the original game by using an "objective chance mechanism" [3].

2. *Computation of Correlated Equilibrium*

Considering an N-player game with multiple actions, the game can no longer be represented in bimatrix form. As presented in Sec. II.A.1, let $S = \{S_1, S_2, \cdots, S_N\}$ be the strategy set of the players, $S_i = \{s_{i_1}, s_{i_2}, s_{i_3}, \cdots, s_{i_{a_i}}\}$ be the set of available actions, and a_i be the total number of actions for the player i. Let the total number of elements in $S_p = S_1 \times S_2 \times \cdots \times S_N$ be n_p defined as

$$
n_p = \prod_{m=1}^{N} a_m \tag{5}
$$

The joint probability P that should be computed has n_p elements, $P = (p_1, p_2, \cdots, p_{n_p})$. Now to construct the constraint inequalities, the following steps are involved. Assume that player 1 is recommended to play its first action, s_{1_1}. The expected payoff for player 1 is

$$
\begin{aligned}
\mathrm{E}U(1) = {}& p_1 U^1_{1_1,2_1,3_1,\cdots,N_1} + p_2 U^1_{1_1,2_2,3_1,\cdots,N_1}\\
&+ \cdots + p_k U^1_{1_1,2_{a_2},3_{a_3},\cdots,N_{a_N}}
\end{aligned}
\tag{6}
$$

where p_k is the probability corresponding to the last strategy distribution (which corresponds to player 1 using its action 1 and rest using their a_ith action) and U^1 is the payoff for the corresponding strategies. Now, player 1 could deviate from the recommendation and use any of its a_1 actions. The expected payoff for using action j is

$$
\begin{aligned}
\mathrm{E}U(1) = {}& p_1 U^1_{1_j,2_1,3_1,\cdots,N_1} + p_2 U^1_{1_j,2_2,3_1,\cdots,N_1}\\
&+ \cdots + p_k U^1_{1_j,2_{a_2},3_{a_3},\cdots,N_{a_N}}
\end{aligned}
\tag{7}
$$

where the distribution is that for the recommended action 1. Now substituting these in Eq. (1),

$$
\begin{aligned}
& p_1 (U^1_{1_1,2_1,3_1,\cdots,N_1} - U^1_{1_j,2_1,3_1,\cdots,N_1})\\
&\quad + p_2 (U^1_{1_1,2_2,3_1,\cdots,N_1} - U^1_{1_j,2_2,3_1,\cdots,N_1})\\
&\quad + \cdots + p_k (U^1_{1_1,2_{a_2},3_{a_3},\cdots,N_{a_N}} - U^1_{1_j,2_{a_2},3_{a_3},\cdots,N_{a_N}}) \geq 0
\end{aligned}
\tag{8}
$$

Given a recommendation to play one particular action out of a_1 actions available, the player 1 can use $a_1 - 1$ other actions instead of the recommended one. For each of the a_1 actions in its set, player 1 has the same number of options to deviate. Thus, there are $a_1(a_1 - 1)$ ways of choosing another action when a recommendation is given to player 1. A similar computation for other players would give the total number of deviations that each player could carry out given a recommendation. This is equivalent to n_{ineq}, the number of inequality constraints:

$$
n_{\text{ineq}} = \sum_{m=1}^{N} a_m (a_m - 1) \tag{9}
$$

with n_p variables. The bounds on the variables are expressed as

$$
\begin{aligned}
&\sum_{m=1}^{n_p} p_m = 1\\
&\quad p_m \geq 0, \quad m = 1, 2, \cdots, n_p
\end{aligned}
\tag{10}
$$

This can now be formulated as a linear program with an objective function represented as

$$J = \sum_{m=1}^{n_p} b_m p_m \tag{11}$$

where b_m are the weights. The constraint set is

$$[A]_{n_{\text{ineq}} \times n_p} \le 0 \tag{12}$$

where A represents the n_{ineq} inequalities in n_p variables, computed as in Eq. (8). This, along with the equation and bounds given in Eq. (10), constitutes the linear programming problem to solve for the correlated equilibrium. The correlated strategies are those that satisfy the constraints, whereas the equilibrium is the one that optimizes the objective.

The above formulation can be demonstrated by an example. Consider a game with $N = 3$ and total number of actions $a = \{2, 1, 3\}$. Here, the strategy set S_p has $n_p = 6$ elements and the number of inequality constraints is $n_{\text{ineq}} = 8$. Probability P to be computed is $P = (p_1, p_2, p_3, p_4, p_5, p_6)$. The notation for payoff is modified here for simplicity as $U^x_{x_1,x_2,x_3}$, the payoff of player x when the strategy used is $\{x_1, x_2, x_3\}$. The constraints for correlated strategies are given below.

$$\begin{aligned}
&p_1(U^1_{1,1,1} - U^1_{2,1,1}) + p_2(U^1_{1,1,2} - U^1_{2,1,2}) \\
&\qquad + p_3(U^1_{1,1,3} - U^1_{2,1,3}) \ge 0 \\
&p_4(U^1_{2,1,1} - U^1_{1,1,1}) + p_5(U^1_{2,1,2} - U^1_{1,1,2}) \\
&\qquad + p_6(U^1_{2,1,3} - U^1_{1,1,3}) \ge 0 \\
&p_1(U^3_{1,1,1} - U^3_{1,1,2}) + p_4(U^3_{2,1,1} - U^3_{2,1,2}) \ge 0 \\
&p_2(U^3_{1,1,2} - U^3_{1,1,1}) + p_5(U^3_{2,1,2} - U^3_{2,1,1}) \ge 0 \\
&p_1(U^3_{1,1,1} - U^3_{1,1,3}) + p_4(U^3_{2,1,1} - U^3_{2,1,3}) \ge 0 \\
&p_3(U^3_{1,1,3} - U^3_{1,1,1}) + p_6(U^3_{2,1,3} - U^3_{2,1,1}) \ge 0 \\
&p_2(U^3_{1,1,2} - U^3_{1,1,3}) + p_5(U^3_{2,1,2} - U^3_{2,1,3}) \ge 0 \\
&p_3(U^3_{1,1,3} - U^3_{1,1,2}) + p_6(U^3_{2,1,3} - U^3_{2,1,2}) \ge 0 \\
&\qquad\qquad p_l \ge 0 \; l = 1, \cdots, 6 \\
&p_1 + p_2 + p_3 + p_4 + p_5 + p_6 = 1
\end{aligned} \tag{13}$$

An appropriate objective function could be defined to find the equilibrium strategy P that satisfies the constraints in Eq. (13). The expected payoff for the player m is given below.

$$\begin{aligned}
\mathrm{E}U^m = {} & p_1 U^m_{1,1,1} + p_2 U^m_{1,1,2} + p_3 U^m_{1,1,3} + p_4 U^m_{2,1,1} \\
& + p_5 U^m_{2,1,2} + p_6 U^m_{2,1,3}; m = 1, 2, 3
\end{aligned} \tag{14}$$

Correlated equilibrium is computationally efficient but it may become intractable with increasing (n_p). The value of n_p depends upon number of players and number of actions chosen by them. In a regulated airspace, it is unlikely to have a very large number of vehicles in conflict. A judicious choice of number of actions for each player can limit the increase of n_p.

B. Collision Avoidance

As discussed in Sec. I, different methods could be employed for local path planning when a collision threat is detected by a vehicle. An efficient avoidance algorithm should be computationally light and have a quick response with minimum information or sensing capability. It should be preferably optimal in possible aspects such as avoidance time and power requirement. Figure 2 shows the on-board system architecture of a typical UAV. The guidance and control modules are integrated into the vehicle to access the internal parameters of the UAV via the sensor suite and compute necessary commands for a smooth flight. The on-board system sends state information of the vehicle to the ground monitoring station via a communication module. It receives information from the ground stations as well as other vehicles within range. The radio frequency receiver allows the pilots to have a manual override and take over the UAV control in case of faults. The conflict detection and resolution modules take care of the collisions that might occur in the vehicle's path. They predict incoming threats using sensor feed and recompute the vehicle's trajectory to reach its destination safely.

The avoidance problem changes significantly according to the nature of the response to the incoming threats. When it comes to interactions among multiple agents, a broad classification would be cooperative and non-cooperative agents. The cooperative agents fully share information and mostly have a common goal. Such agents are owned mainly by a single party or are deployed for a particular mission. The non-cooperative agents, on the other hand, strive to achieve their own goals with customized preferences. This behavior may be either due to their belonging to different groups or due to the absence of a mechanism to cooperate with others. In all these situations, safe travel needs to be ensured. Incoming unknown threats could be detected by on-board sensors like Light Detection and Ranging (LIDAR), ultrasound sensor, camera, whereas sensors like Automatic Dependent Surveillance-Broadcast (ADS-B) help in cooperative conflict detection and prediction. However, only those obstacles that are within the range and field of view (FoV) of these sensors could be detected. Carrying long-range and wide-FoV sensors comes at the cost of higher power consumption and reduced flight time. Thus, detection relying on on-board sensors alone might not confirm safety. Because ensuring safety among cooperative agents itself is difficult, due to reasons like computational constraints, sensor limitations, and unreliable communication systems, ensuring non-cooperative agents' safety is a more complicated task.

Considering the above factors, UAVs' deployment into the airspace in large numbers requires careful planning in various directions. This

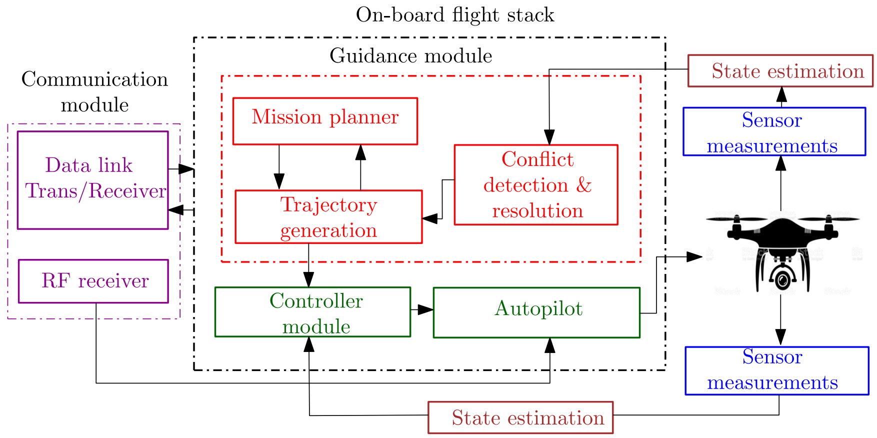


Fig. 2 Block schematic of a UAV system architecture.

would involve developing the physical infrastructure of UAV hardware, ports, and communication systems to software solutions for traffic management and routing of UAVs. Regarding the software solutions, this work proposes a novel semi-autonomous mechanism for conflict resolution in traffic management. Implementation of such a semi-autonomous system would enable a better understanding of the complete system and possibilities of its improvement while reducing any chances of catastrophic events due to the deployment of a fully autonomous system.

The following section gives the details of CONCORD system, the proposed UAV conflict resolution system that uses the concept of correlated equilibrium.

III. Decision Making Using CONCORD

When a large number of vehicles use airspace, the primary factor of concern is their safety. Collision avoidance is an essential and unavoidable feature for all autonomous vehicles. The ATC manages traffic to ensure conflict-free 4D-T trajectories for the aircraft [49]. Loss of separation is whenever a horizontal or vertical minimum separation is breached. Termed as AIRPROX by ICAO [50], the situation is taken care of by ACAS [51], where the pilot is provided assistance in the form of advisories. A traffic advisory is made first to give a visual assistance on the conflicting vehicle and a resolution advisory to help control the aircraft out of the situation. A similar architecture could be implemented for unmanned vehicles, where the conflict resolution decisions are mediated from the ground station. This work proposes a novel architecture in which the decision making associated with vehicles' deconfliction is carried out using the concept of correlated equilibrium. The formulation of the avoidance problem as a game and its resolution by finding the correlated equilibrium has the following benefits. The players' deviation from the computed solution will naturally bring them back to the computed solution itself due to the equilibrium point's stability property. This means that when vehicles execute avoidance not following the computed solution, a lower payoff results. The reduced payoff forces the vehicles to follow the computed solution in the next conflict instant. Thus, the expected utility values of vehicles converge to an optimal value upon repeated conflict resolutions.

The state-of-the-art traffic management systems introduce UAVs in an organized fashion into the airspace, rather than a highly disordered group of vehicles taking care of individual goals. This also means that the vehicles' global path planning could take care of any obvious conflict risks. Nevertheless, the safety issues are not entirely resolved. Several factors affect the precomputed conflict-free flight path, like the uncertainties and external disturbances, and noises. In such scenarios, careful rerouting of the UAVs is vital to avoid crashes.

A. CONCORD System

The vehicles may or may not cooperate. Communication with a ground control station is assumed mandatory for the vehicles. The block schematic of the proposed CONCORD system is given in Fig. 3. The UAVs traverse in the regulated airspace and transmit their state information (such as position and velocity) periodically to the control station. It is assumed that a suitable communication infrastructure exists to take care of this information exchange. CONCORD would be the conflict resolution subsystem for a typical semi-autonomous UAV traffic management system, consisting of other subsystems like UAV monitoring, scheduling, routing, and geofencing permit-to-fly. The vehicles employ their avoidance module to compute the resolutions or rely on the control station for resolution advice. This is a realistic assumption that allows the vehicles to propose resolutions without violating their physical constraints and priorities and reducing the control station's load. When a UAV takes off for a mission, it registers its information with CONCORD. This information includes the priority metric associated with them. The priority could be to optimize energy, time, or any other quantity. This quantity would ideally represent the priority assigned to the operator that has deployed the particular UAV. Therefore, the ground station has the utility information of the vehicles. In a conflict scenario, the UAVs may not compute resolutions considering individual priorities of the vehicles in conflict. But CONCORD computes the correlated avoidance strategies using the shared states and already known utility functions. This ensures that the individual vehicle priorities are considered while computing avoidance control. Thus, the UAVs do not share information about their utilities online but are sent to the base station at the beginning of their missions.

Although a centralized system is not a robust option for conflict resolution, it is definitely a step toward a fully automated and distributed avoidance framework in a heterogeneous UAV airspace. CONCORD mediates to ensure fair avoidance among UAVs. In the

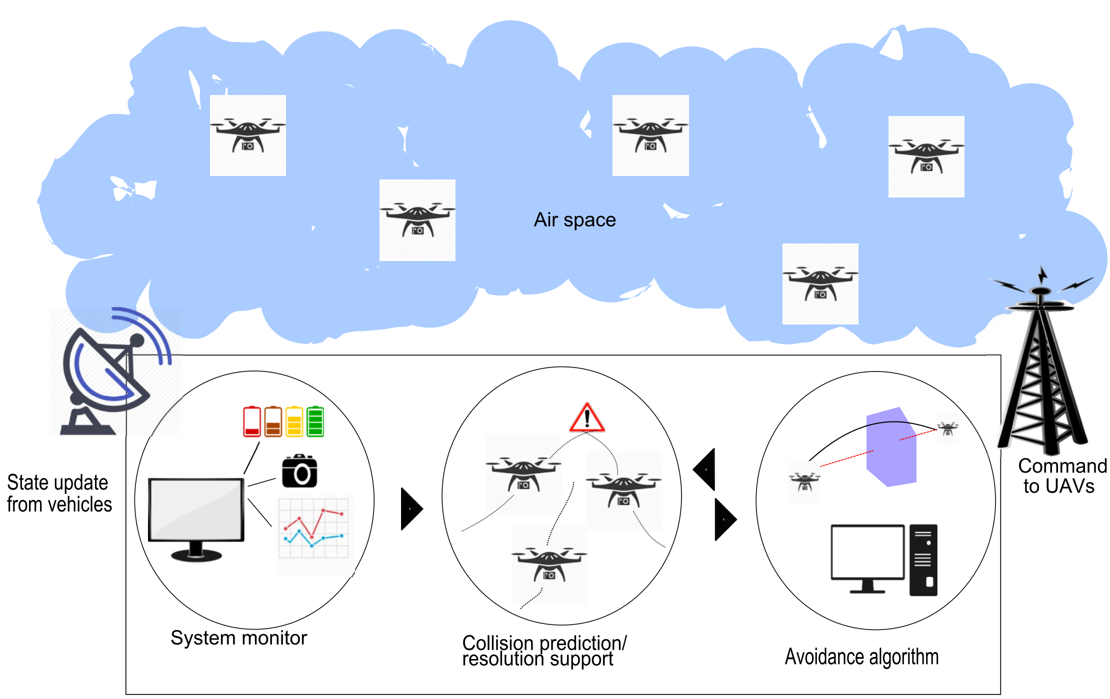


Fig. 3 Block schematic of the proposed CONCORD system.

proposed work, we have used linear programming to compute the correlated equilibria. However, unlike Nash equilibrium, learning methods could be employed [52] to compute correlated equilibrium.

B. Resolution Among Two UAVs

Consider a pair of vehicles A and B moving from points A_0 and B_0 to their respective goal locations A_T and B_T, as shown in Fig. 4a. At some instant of their travel, the control station predicts that there occurs a loss of separation and hence a conflict. Upon conflict prediction, control is transferred to the CONCORD system.

1. Cooperative Avoidance via CONCORD

Here, the vehicles' priorities and the parameters they would like to optimize are known to CONCORD. In terms of game theory, this setup could be thought of as players who have an externally enforced binding agreement with other players in the airspace it shares. The vehicles receive intruder information and compute their respective resolutions. These are transmitted to the control station where CONCORD solves for the correlated strategy using these. The strategies are transmitted back to the vehicles. The vehicles execute the recommended action to avoid collision and proceeds to their respective goal locations. The process execution by the CONCORD and UAVs for cooperative avoidance is as follows.

The vehicles transmit their state information every Δt s, which is the update rate of the system. The mission duration is represented by

$$t_{\mathrm{MA}} = t_{\mathrm{gA}} - t_{\mathrm{sA}} \tag{15}$$

$$t_{\mathrm{MB}} = t_{\mathrm{gB}} - t_{\mathrm{sB}} \tag{16}$$

where t_{MA} and t_{MB} are the mission duration of each vehicle, t_{sA} and t_{sB} are the starting instants, and t_{gA} and t_{gB} are the computed time instants to reach the goal location for UAVs A and B, respectively. As the vehicles proceed, at $t = t_{\mathrm{Pr}}$, a conflict is predicted at t_c s. These instants of conflict prediction are marked in Fig. 4a as Pr_A and Pr_B and instants of conflict resolution as R_A and R_B, for UAVs A and B, respectively. The vehicles are on collision course if the vehicle to vehicle separation is

$$r_{\mathrm{v2v}}(t) = \|X_i(t) - X_j(t)\|_2 \le r^d_{\mathrm{v2v}_{ij}} \tag{17}$$

where X_i and $X_j \in \mathrm{R}^3$ are instantaneous positions of UAVs i and j and $r^d_{\mathrm{v2v}_{ij}}$ is the desired minimum separation between the pair of UAVs, which is assumed to be predefined.

Region of safety around each vehicle is considered to be spherical, with a radius r^d_v, as shown in Fig. 5. Conflict occurs if the spherical

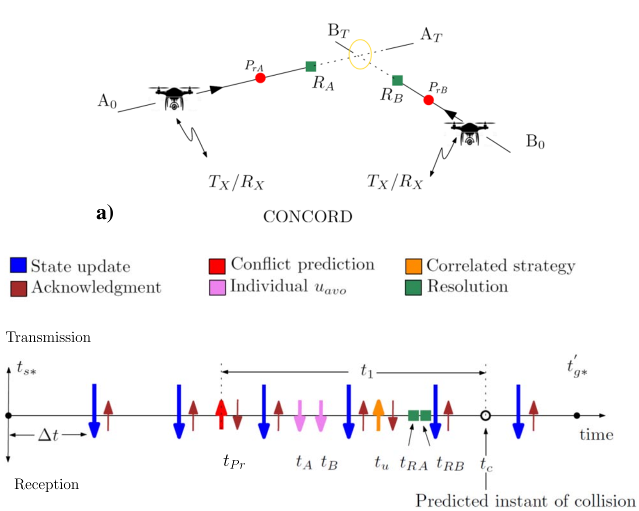


Fig. 4 a) Cooperative avoidance scenario. b) Communication timeline.

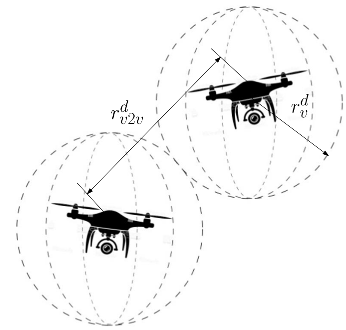


Fig. 5 Minimum inter-UAV and UAV safe distance.

volume around the UAV of radius r^d_v is breached. In literature, "well clear" [53–56] for UAVs is defined as maintaining horizontal and vertical separation between vehicles. This would result in a safe cylindrical volume. In this work, we use a spherical volume of safety. This can be extended to other definitions as well. The conflict warning message would consist of the number of intruders (one in this case) and state information of the intruders and time to minimum separation, t_{ms}. Time to minimum separation is the time remaining for collision from the instant of prediction, which is t_1 here. As a response to this, the vehicles compute their recommendations u_{avo} and transmit this information to the CONCORD system. Let the recommendations from UAVs A and B be received by CONCORD system at t_A and t_B, respectively, as shown in Fig. 4b. The values of t_A and t_B would be upper bounded based on t_{ms}, speed of computation for decision making, communication delays, etc. CONCORD computes the correlated strategies for the vehicles and transmit at t_u and the vehicles acknowledge receipt for confirmation. The resolution essentially consists of the maneuver to be applied and time at which the vehicles should start executing the maneuver. The conflict resolution begins at t_{RA} and t_{RB}, for UAVs A and B, respectively. Subsequently, the time of mission is modified to

$$t_{M*} = t'_{g*} - t_{s*} \tag{18}$$

The mission time t_M is an important aspect when it comes to permissions and availability of the airspace being used. It should be noted that any delay in the transmission and reception of information is independent of the CONCORD module. Also, the resolution is executed at two different instants of time (t_{RA} and t_{RB}), which may also occur simultaneously, or in a different order other than the one shown here.

2. Non-Cooperative Avoidance via CONCORD

The case of non-cooperating UAVs is interesting. They cover a range of realistic possibilities like vehicle sensor damage, issues with communication, or even the case of high-priority vehicle that cannot deviate from its preplanned mission. In such scenarios, the vehicles' safety depends on the responsive UAV. In a non-cooperative setting, the role of CONCORD is even more critical. The decision making and maneuvering in an entirely autonomous setup do not happen, due to which the vehicles may breach each other's safety margins. The idea here is to compute the best response to the anticipated action of the non-cooperative vehicle. Unless it is a hostile or rogue vehicle, it is quite likely for the non-cooperative UAV to maintain its current path without deviating. This behavior is the anticipated response and is captured in the computation of correlated strategies for non-cooperative UAVs.

Consider the same scenario in Fig. 4a. Similar to the cooperative setup, conflict is predicted in the first stage. However, the control station identifies the non-cooperative nature of one of the vehicles. This identification could be by explicit messages (high-priority crossing cases) or unresponsiveness of the vehicle (damage or any other reason). The communication timeline shown in Fig. 6 consolidates the entire procedure executed by CONCORD. Here, in contrast to the cooperative scenario, UAV B is assumed to be non-cooperative. It

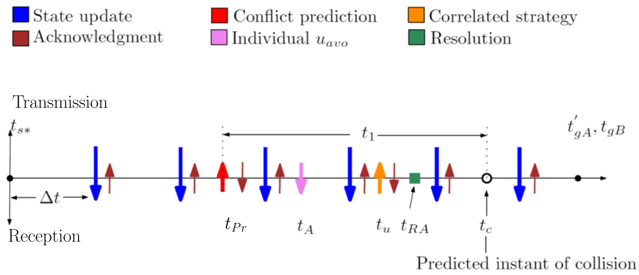


Fig. 6 Communication timeline in case of non-cooperative UAVs.

broadcasts the state information but does not respond to the conflict warning from the control station. Due to this behavior, the CONCORD requests UAV A for avoidance recommendations. The correlated strategy is computed and transmitted to UAV A, assuming that UAV B does not change its course. A new prediction is made if it does, and a second conflict warning is issued to UAV A with the new intruder parameters. In scenarios where the time remaining for conflict is less, CONCORD signals to initiate a fail-safe mode. If UAV B maintains its broadcast states, a successful conflict resolution would be achieved. The time of mission for UAV B would remain the same while that of UAV A would be $t'_{\mathrm{gA}} - t_{\mathrm{sA}}$.

C. Role of Correlated Equilibrium in CONCORD

The correlated equilibrium occurs when the players follow the recommendations from the mediator or correlation device. This requires trust between the players and mediation entity. For non-cooperative players, CONCORD system does not compute resolutions. Instead, based on the estimated state information of non-cooperative UAVs, avoidance resolutions are computed for the other vehicles. The other cooperative UAVs follow the recommendation due to their trust in CONCORD system. CONCORD operates to resolve conflicts by considering priorities that can be translated to utility functions and does not favor any specific parties. Thus, trust between vehicles and CONCORD can be ensured.

The CONCORD module continuously monitors and receives information of the vehicles. In case a conflict is predicted, it computes resolutions for the players. The players need not know the whole game or any part of it in this case. If the entire game is to be shared with all the players, this would result in requirement of superior/homogeneous memory and computational capabilities for all the UAVs. Deconfliction by the central system removes the necessity of sharing the game with the players as the decision-making entity has the information about the whole game.

The CONCORD deconfliction is agnostic to the avoidance module and the kind of avoidance control input. The players get recommendations from the avoidance control inputs modeled in the game and they use them to remain safe.

D. CONCORD for Multiple UAVs

Multivehicle conflict scenarios in unmanned vehicle traffic systems would be a regular event in the future. Many avoidance algorithms are available [57] tailor-made for multi-UAV conflict resolution. The mission duration is divided into smaller time horizons. CONCORD predicts conflicts, if any, at the beginning of each time horizon. The UAVs could compute resolutions and transmit them to CONCORD. The control station could resolve the situation if the individual vehicles fail to work out a solution within a prescribed time window. Whether it is cooperative, non-cooperative, or multivehicle, the nature of conflict is checked before resuming the resolution process. Based on the vehicles' avoidance recommendations or by computing on its own, CONCORD system generates the correlated resolutions for the UAVs. These equilibrium avoidance commands are sent back to the UAVs.

Algorithm 1: Conflict resolution using CONCORD

1 Update the number of UAVs in airspace, N, initialize $N'_c = 0$;
2 Conflicts unresolved in previous time horizon, N_c^{old};
3 State update of UAVs online, $X_i(t_0)$;
4 Check for imminent conflicts N_c^{new} in time horizon t_H;
5 **if** *conflict* **then**
6   Compute t_{ms} and prepare priority list;
7   **while** $(N_c = N_c^{\mathrm{old}} + N_c^{\mathrm{new}} + N'_c) > 0$ **do**
8   **if** *simultaneous conflicts* **then**
9   Select k_c conflicts with the least t_{ms} (high priority);
10   Retrieve information on vehicles (states, cooperative or not, objective);
11   Obtain resolutions from vehicles or compute the same;
12   Compute correlated strategies for vehicles;
13       $N_c = N_c - k_c$;
14   **else**
15   Retrieve information on vehicle (states, cooperative or not, objective);
16   Obtain or compute resolutions for pair of vehicles with least t_{ms};
17   Compute correlated strategy;
18       $N_c = N_c - 1$;
19   **end**
20   Send recommendations;
21   Check for new conflicts, N'_c, if any;
22   **if** $N'_c \neq 0$ **then**
23       $N_c = N_c + N'_c$;
24   Go to 6;
25   **end**
26   **if** $t - t_0 > t_H$ **then**
27   Go to 1;
28   **end**
29   **end**
30 **else**
31   Receive state update;
32   Send acknowledgment;
33 **end**

The communication timeline is similar to the ones described above, except for the increased number of transmissions and receptions among multiple vehicles. The underlying algorithm of CONCORD is given in Algorithm 1. For N UAVs using the airspace, conflicts are predicted within the time horizon t_H. Based on the time to minimum separation, t_{ms}, and their locations, it is classified into simultaneous and pairwise conflicts. Simultaneous conflicts correspond to multiple conflicts occurring at the same time within a time horizon. Pairwise conflicts only involve a pair of UAVs.

N_c refers to number of vehicles in conflict. After each iteration, CONCORD checks for new conflicts N_c^{new}. This new number of conflicts is added to the existing number before beginning the next iteration. The proposed algorithm records the number of conflicts within a particular time horizon (represented by N_C in Algorithm 1) and resolves it based on least time to conflict. However, there is a possibility of number of conflicts increasing. High-density traffic might result in this cascading effect. This is unlikely to happen in practice. If this happens, the algorithm can take care of the same in the next time horizon. If the conflict is predicted to occur outside the current time horizon, the conflict is carried over to the next horizon for resolution (represented by N_c^{old} in Algorithm 1). If not, it will be resolved in the current horizon itself.

The time horizon t_H is a fixed time window within which conflicts are checked for and resolved. The time horizon is fixed depending on the time required to resolve conflict by CONCORD, system delays, and UAV response times. There could occur a chain of conflicts over time, which the CONCORD system can take care of. However, it seems practically impossible for all vehicles in an airspace to be in conflict with each other as the CONCORD framework is designed envisaging a structured UAV traffic rather than a group of UAVs with motion in random directions. Thus, N_c should be bounded under this assumption.

The algorithm is designed envisaging a structured UAV traffic rather than a randomly moving group of vehicles. For the former, it is quite unlikely for all N vehicles in an airspace to be in conflict simultaneously. Therefore, dominance of ranking t_{ms} in step 6 of the algorithm is unlikely to happen in practice.

E. Computational Complexity

In this section, the computational complexity of Algorithm 1 is being evaluated. Let the time required for step 1 be T_1 units, step 2 be T_2 units, and, similarly, step i be T_i units. Steps 1–5 are run once. The time required for executing them is

$$T = T_1 + T_2 + T_3 + T_4 + T_5$$

Step 4 has complexity of order $O(N^2)$, as it requires comparison between N UAVs in worst case.

Step 6 evaluates the time to minimum separation for the UAVs in conflict and sorts them. This operation has a complexity of order $O(N_c) + O(N_c \log N_c)$. This is evaluated only once if no new conflicts are detected within the current time horizon. If not, step 6 will be evaluated d times, where d is the number of times new conflicts are detected within a time horizon. The "while" loop is evaluated N_c times. The conflict is identified as simultaneous or pairwise. The conflicts are simultaneous if multiple UAVs are involved in conflicts, leading to a single point engagement or multiple engagements. Otherwise the resolution is computed pairwise.

For the first case, let there be m simultaneous conflicts. So the "If" part is evaluated m times. Time required for the evaluation is

$$T_{\mathrm{If}} = m(T_8 + T_9 + T_{10} + T_{11} + T_{12} + T_{13})$$

For the latter, let there be n pairwise conflicts such that $N_c = m + n$. So, the "else" part is evaluated n times. The time required is

$$T_{\mathrm{else}} = n(T_{14} + T_{15} + T_{16} + T_{17} + T_{18})$$

Steps 20 and 21 are evaluated mn times, which requires mnT_{20} units of time.

Steps 22–24 are evaluated d times, following the logic given earlier. Step 26 is evaluated mn time, whereas step 27 is evaluated once. Steps 31 and 32 are evaluated once. The total time required for the evaluation of the algorithm is

$$\begin{aligned}
T_{\mathrm{total}} &= T_1 + T_2 + T_3 + T_4 + T_5 + dT_6 + (N_c + N_c')T_7 + m(T_8 + T_9 \\
&\quad + T_{10} + T_{11} + T_{12} + T_{13}) + n(T_{14} + T_{15} + T_{16} + T_{17} + T_{18}) \\
&\quad + mnT_{20} + d(T_{22} + T_{23} + T_{24}) + mnT_{26} + T_{27} + T_{31} + T_{32} \\
&= C_0 + O(N_c) + O(N_c \log N_c) + d(T_6 + T_{22} + T_{23} + T_{24}) \\
&\quad + m(C_1 + T_{11} + T_{12}) + n(C_2 + T_{16} + T_{17}) + mn(C_3 + C_6)
\end{aligned}$$

Here, T_{12} and T_{17} are the evaluation times for computing correlated strategies, which is done using a linear programming formulation involving n_p variables. For ellipsoid algorithm, the complexity is polynomial P. T_{11} and T_{16} are the time required for conflict resolution, whose complexity is taken as $O(l_u^2 l_t)$ [20]. Therefore, the time complexity of the CONCORD system in resolving conflicts for one time horizon is

$$\begin{aligned}
T_{\mathrm{total}} &= C_0 + O(N_c) + O(N_c \log N_c) + d(C_7) + m(C_1 \\
&\quad + O(l_u^2 l_t) + n_p^6 C_4 + C_5) + n(C_2 + O(l_u^2 l_t) + n_p^6 C_4 \\
&\quad + C_5) + mn(C_3 + C_6)
\end{aligned}$$

Complexity is of the order $O(N_c + N_c \log N_c + (m+n) l_u^2 l_t + (m+n) n_p^6 + mn + d)$. This implies that the complexity of the algorithm is dependent on the number of variables n_p, which is the product of number of resolutions of the UAVs. It also depends on number of multivehicle resolutions, m, and number of pairwise conflicts, n. It should be noted that the complexity for the linear program considered here is that for the ellipsoid algorithm. Any time optimal algorithm could be used and the complexity would be optimized accordingly. This complexity is feasible in comparison with the Nash equilibrium complexity that is NP-hard. A large number of resolutions considered would result in a higher value of n_p^6 and the complexity may increase. Thus, a judicious choice on number of actions for the players is recommended for better run time performance.

IV. Integration of CONCORD into UAV Traffic Management System

This section describes how CONCORD could be integrated into an existing or future Unmanned Aerial Vehicle–Traffic Management System (UAV-TMS).

The intent of this section is not to define a new traffic management system, but to show how the CONCORD system could be a part of any available UAV traffic management framework as a conflict resolution module. The conflict resolution model of UTM has preflight and postflight avoidance approaches [58]. Tactical and strategic avoidance approaches are proposed, but there is no inclusion of fairness metric in computing avoidance actions. CONCORD, being a decision-making module for fair deconfliction, will add value to an existing UAV traffic management system. Several other advantages of correlated equilibrium discussed in Sec. II make it a good choice for conflict resolution. These advantages are being computationally efficient and giving better payoff to the players, among others. Considering priorities of parties involved in UAV deployment results in safe transit and social/economic satisfaction. The unrestricted airspace would consist of autonomous and remotely piloted UAVs, as shown in Fig. 7. The service providers do the segregation and routing of these vehicles. They are the immediate authorities who take care of all aspects of the UAV mission.

Ensuring the safety of not just the vehicles but also the human population below is of prime importance. Before the flight, all necessary checks would be carried out to confirm this. However, there are several factors, internal and external to the vehicles, that might lead to conflict between the UAVs while in flight. Examples of these are noisy sensor measurements, external disturbances like wind and gust, and unanticipated speed changes during flight. Such situations may not be rare, and hence relying on the vehicle alone to resolve based on the state-of-art technology may complicate the situation. Thus, such in-flight conflicts could be solved better by integrating the CONCORD subsystem to the UAV-TMS.

The block schematic giving the details about integration of CONCORD to an existing UAV-TMS is shown in Fig. 8. Traffic management and control are hierarchical with a top layer, managing UAV service providers and coordinating with ATC. Its primary functions include authentication, data collection and analysis, flight data management and operator management, and security. The middle layer consists of operators or service providers who monitor, authorize, and regulate the UAV traffic for different applications. They do so by dedicated software setup, communicating with the users, and scheduling services according to the demand. UAV service demands are diverse, as mentioned in Sec. I. This requires the UAV-TMS to coordinate among different service providers to expedite the regulatory aspects and ensure smooth traffic management. The service providers may or may not communicate with each other. Nevertheless, flight data are updated on a central server, enabling multiple service providers to work together in case of heavy traffic and service demand. The collaborating parties could do airspace sharing at the operator level at the middle layer. This reduces the load on the top level from sanctioning multiple missions in overlapping regions. The operators also ensure a sufficient number of pilots to take manual control in any unexpected event.

The lowermost layer is the users who would require the deployment of UAVs for various applications. They interact with the UAV-TMS via the service provider user interface. CONCORD being integrated into the ground control station ensures vehicle safety. The flight data are communicated to the ground control station for monitoring and regulatory purposes. The CONCORD system uses

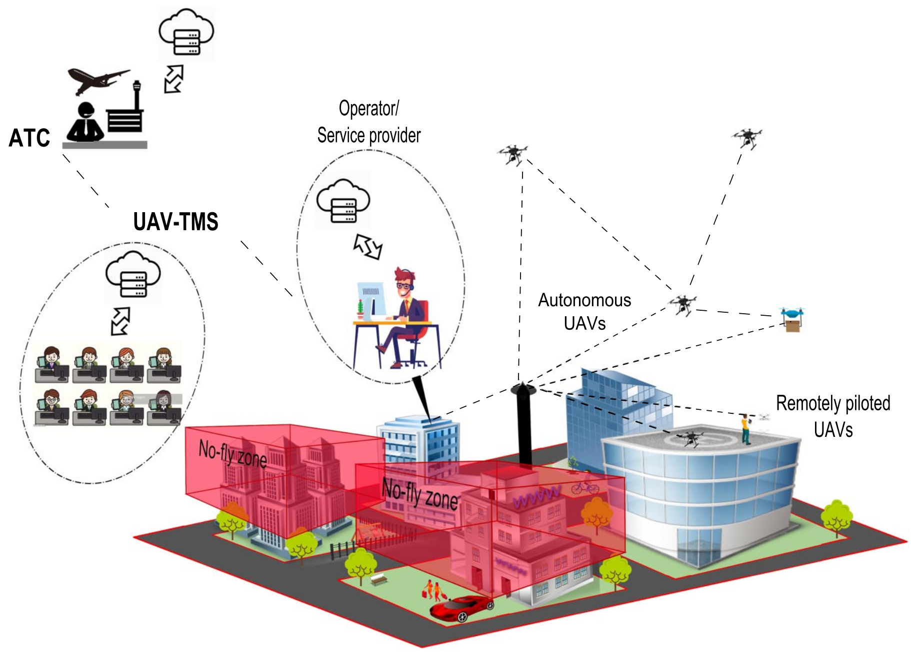


Fig. 7 Schematic of UAV-TMS in an urban setup.

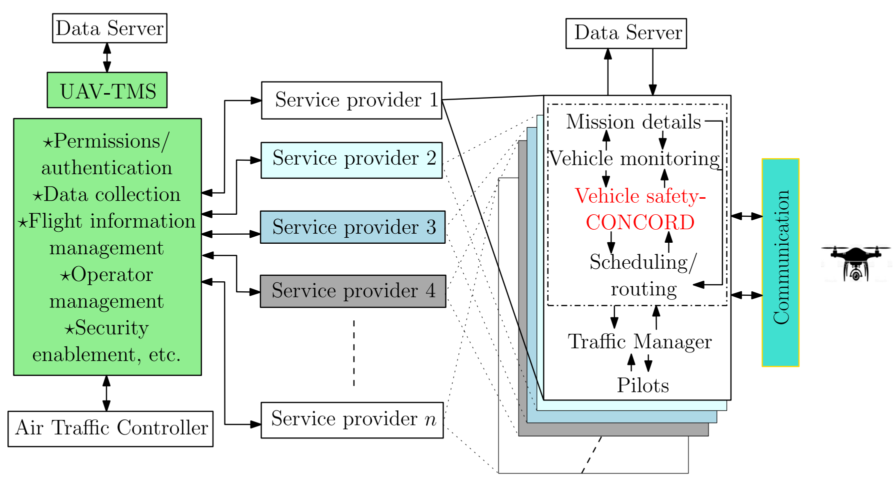


Fig. 8 Block schematic of UAV-TMS and CONCORD integration.

this information to predict conflicts. Because the conflicts are detected during the flight, it is crucial to have a time-bound resolution that considers the priorities of all the involved UAVs. CONCORD qualifies for this task as a decision-making module that considers the utilities that the individual vehicles would prefer to optimize reasonably with less computational complexity.

V. Numerical Simulations and Examples

In this section, we present few examples to demonstrate the CONCORD system. Cooperative, non-cooperative, and multiple UAV scenarios are considered along with a use case of a typical structured traffic. The models considered are for simplicity. The concept could be applied to any realistic system because CONCORD is a decision-making module that could be integrated with any traffic management system and with any collision avoidance algorithms. The simulations are carried out in MATLAB 2015.b and the processor used is Intel CORE i5, 10th gen with 8 GB RAM. This section consists of the following scenarios. At first, a toy example is considered. Then two UAVs traveling in intersecting paths in repeated fashion are presented. Three UAVs simultaneously reaching an intersection and hence resulting in conflict are given subsequently. The last simulation scenario considers an urban environment with structured traffic where UAVs travel through predetermined paths and deconfliction is carried out by CONCORD system. Randomization of vehicle speeds and Monte Carlo simulation of the resulting scenario are also presented for the structured traffic environment. This demonstrates how CONCORD system ensures fair deconfliction. Each of these cases represents a different scenario, and the examples demonstrate how CONCORD deals with them. These scenarios are individually considered here as it would be unlikely that all of them will occur in a practical situation, and to devise a simulation that would make this happen would require us to contrive a set of initial conditions. However, it is important that the user has the confidence that CONCORD will function well under these conditions, should they arise.

1. Two UAVs in Conflict

This example demonstrates how symmetric nature of payoff matrix affects the equilibrium payoff. For a pair of UAV, three scenarios are considered. Each UAV has three actions in the first case and five actions in the second. The third scenario has three actions for each UAV but in windy environment.

Consider the example shown in Fig. 9. Vehicles 1 and 2 are on a collision course and they do not communicate among each other. The players are selfish in that they would like to maximize their payoff.

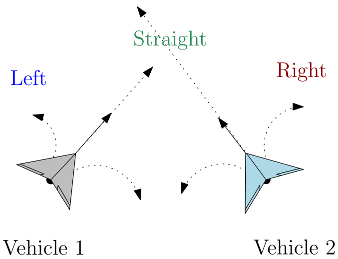


Fig. 9 Vehicles in collision course with chosen action set.

We assume that the avoidance scheme used generates three options, namely, left, straight, and right. Although, in reality, the UAVs will have more options such as graded left and right turns, and also options of going up or down, we consider only three options for the purpose of illustration. For each of these actions, the vehicles get some payoff that could be interpreted as their penalty/reward for collision or avoidance. A representative payoff matrix is given in Table 3.

The payoff values should be reasoned as follows: When both travel straight, both collide, gaining negative payoff. When one goes straight and other takes a turn, collision is avoided but the one that went straight expended no effort, which gives it higher payoff than the one that turned. When both turn, collision is avoided, and both have equal payoff. All action pairs leading to collision are given negative payoff. The Nash equilibria for the game are given in Table 4. The pure strategies could be anyone from the four pairs. There are three mixed strategies as well (to be read as Vehicle 1 mixing S and L with 0.5 probability each, and, similarly, Vehicle 2 mixing S and L with 0.5 probability each). For these, when one player achieves higher payoff, the other gains nothing. Because the players do not communicate with each other, achieving Nash equilibria is not straightforward as well. It should be noted that the payoff values shown above are not computed using physical quantities but are manually assigned to demonstrate the idea of decision making using correlated equilibrium in a two-player game.

Having a mediator makes the problem simpler. The mediator could choose an action and communicate to the respective vehicles. The mediator here is the CONCORD system. For the set of actions considered and the associated payoff values in Table 3, the constraints for correlated strategies are formulated. For the two-player three-action game ($N = 2$, $a_1 = a_2 = 3$), the solution space is of dimension 9 (i.e., $n_p = 3 \times 3$ and $P = (p_1, \cdots, p_9)$) with 12 strategic incentive constraints (i.e., $n_{\text{ineq}} = 3 \times 2 + 3 \times 2 = 12$). Now, solving these to optimize the expected payoff results in the set of correlated probability distributions. One such distribution from among the set is given in Table 5.

The values are chosen such that the probabilities corresponding to (S, S) and (R, L) are zero. This implies that the commands computed by the avoidance algorithm are confirmed safe and beneficial for both the players by the decision-making layer when the correlated distributions are computed. With the given distribution, $\mathrm{E}U(V1) = 2.77$ and $\mathrm{E}U(V2) = 2.77$. Thus, the UAVs could be out of collision course by following the recommendations from the CONCORD system with improved benefit. Once the UAVs are out of collision course, an efficient replanning could be done to redirect its path to the target location without ending up in collision course again. Figure 10 shows the convex hull of Nash equilibrium and correlated equilibrium payoffs. The Nash equilibrium payoffs are computed to plot its convex hull, represented by the quadrangle at the center. It represents the set of payoffs achievable via Nash equilibrium. The convex hull of correlated equilibria is obtained by computing the probability distribution that optimizes the convex combination of payoffs. This set is represented by the outer quadrangle including the unshaded one at the center. This set constitutes all the payoff values achievable by correlation of strategies. It can be seen that the convex hull of Nash equilibria is within that of the correlated equilibrium payoffs. The maximum payoff achieved by correlation is (3, 3), which is a better utility for both the players.

If this case had more options, say, altitude change as well (up and down), the payoff matrix could be computed as given in Table 6. Now the game has two players with five choices each. The payoffs are reasoned as in the previous example. The convex hull of correlated and Nash equilibria is shown in Fig. 11. In comparison to the previous example, the correlated strategies have increased and also the fair payoff is now (4, 4).

Now consider the same pair of vehicles, but in a windy condition, as shown in Fig. 12. Assume the same set of actions as considered

Table 3 Payoff matrix for the vehicles

		Vehicle 2 S	Vehicle 2 L	Vehicle 2 R
Vehicle 1	S	−1, −1	5, 0	5, 0
	L	0, 5	4, 4	4, 4
	R	0, 5	−1, −1	4, 4

Table 4 Nash equilibria of the game

Nash equilibria	Strategy (V1, V2)
Pure	(S, L), (S, R), (L, S), (R, S)
Mixed	(S, L), (S, L) : (1/2, 1/2)(1/2, 1/2)
	(S, L), (S, R) : (1/2, 1/2)(1/2, 1/2)
	(S, R), (S, R) : (1/2, 1/2)(1/2, 1/2)

Table 5 Correlated distribution for the game

		Vehicle 2 S	Vehicle 2 L	Vehicle 2 R
Vehicle 1	S	0	0.21	0.2
	L	0.21	0.06	0.06
	R	0.2	0	0.06

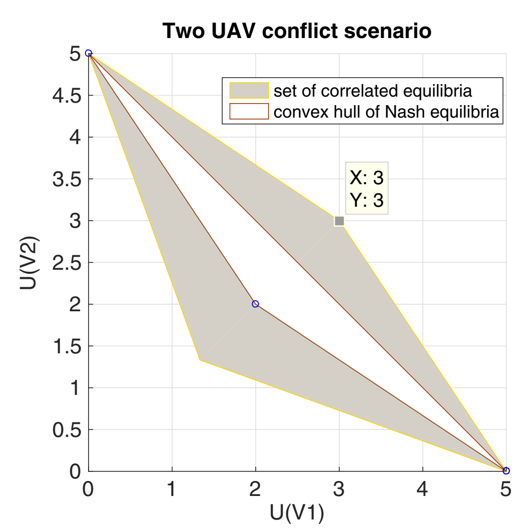


Fig. 10 Convex hull of Nash and correlated equilibria.

Table 6 Payoff matrix for two-player, five-action game

		Vehicle 2 S	Vehicle 2 L	Vehicle 2 R	Vehicle 2 U	Vehicle 2 D
Vehicle 1	S	−1, −1	5, 0	5, 0	4, 4	4, 4
	L	0, 5	4, 4	4, 4	4, 4	4, 4
	R	0, 5	−1, −1	4, 4	4, 4	4, 4
	U	4, 4	4, 4	4, 4	−1, −1	4, 4
	D	4, 4	4, 4	4, 4	4, 4	−1, −1

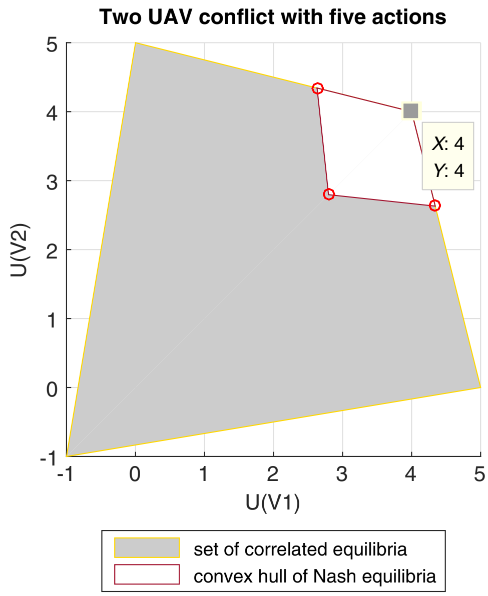


Fig. 11 Convex hulls.

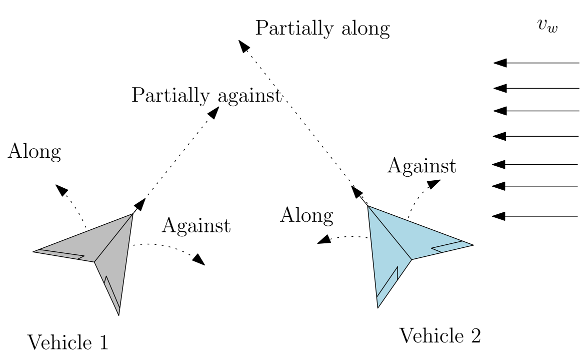


Fig. 12 Vehicles and their avoidance options in a scenario with wind.

before, for both the vehicles. If the feasibility of actions in terms of energy expenditure is considered, each of the action can be classified into four groups (Table 7). This can be reasoned as follows. As shown in Fig. 12, some actions are along the wind (L for Vehicle 1 and Vehicle 2), whereas some are against (R for Vehicles 1 and 2). Action S for Vehicle 1 is partially infeasible because a component of wind is acting against it, whereas S for Vehicle 2 is partially feasible because a component of wind is acting along it. In this case, conflict can be resolved in two ways. The payoffs could be recomputed onboard by the vehicles to find avoidance maneuvers. Another way would be for the ground control station to carry out the computation and then send avoidance maneuvers to the UAVs. The features of semi-automated conflict resolution system could be used here effectively. Recomputing the maneuvers by the vehicles would incur not only computational cost but also time. The payoff matrix corresponding to the ideal environment and its feasibility information could be used by the mediator at the ground control station to provide recommendations. The action feasibility mapped into the payoff matrix is given in Table 8. Those corresponding to favorable actions have two units increment, whereas those unfavorable have two units decrement in payoff. The partial favorable has a unit increment in payoff, whereas the partial unfavorable has a unit decrement. Those leading

Table 7 Actions and their feasibility in windy environment

Vehicle	Favorable	Partially favorable	Partially unfavorable	Unfavorable
1	L	— —	S	R
2	L	S	— —	R

Table 8 Action feasibility mapped to the payoff matrix

		Vehicle 2		
		S	L	R
Vehicle 1	S	−1, −1	4, 2	−24,
	L	2, 6	6, 6	6, 2
	R	−2, 6	−1, −1	2, 2

to collision remains unchanged. The Nash equilibria are (L, S) and (L, L). For this game, the correlated equilibrium is (L, L) and players get an expected payoff of (6, 6). This suggestion prioritizes a fair payoff for both the players and, at the same time, provides an energy efficient resolution.

The payoff matrix in the above examples has a rich symmetrical structure, resulting in a set of correlated strategy corresponding to the quadrangle in the payoff space (Figs. 10 and 11), improving the utility from Nash equilibrium. Even if solutions fail to improve on Nash equilibrium of the game, the advantages of correlation mentioned in Sec. II keep the relevance of CONCORD system intact. There could be cases where the correlated equilibrium would be a point in payoff space, which corresponds to a pure strategy correlated equilibrium. The following examples demonstrate such cases.

2. *Two UAVs in Repeated Conflicts*

This example demonstrates how CONCORD ensures fair deconfliction when repeated conflicts occur between same vehicles. Conflicts are resolved such that the gain/loss for taking avoidance action over multiple runs for the vehicle would be comparable, on an average.

Let us consider the case where two UAVs are in conflict multiple times. Thus, the players are repeatedly playing the resolution game. Assume the following models for the UAVs:

$$\dot{X} = \begin{bmatrix} \cos\theta \\ \sin\theta \\ 0 \end{bmatrix} v + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \omega \tag{19}$$

where $X = [x, y, \theta]$ are the states that represent the position and heading of the UAV, respectively, and v and ω are the linear and angular speed of the UAV, respectively. The examples consider UAVs using avoidance actions on a plane. This consideration is only for simplicity and does not limit the proposed framework because the CONCORD system works with any available avoidance algorithm. The proposed setup can handle avoidance actions in 2D and 3D.

Consider a case shown in Fig. 13a. UAVs A and B start from two locations marked by circles with speeds 20 and 19 m/s, respectively. Their mission is to visit their respective target locations, marked by square, and return to the starting points. The UAVs prioritize to minimize energy expenditure in case of conflicts. The desired waiting time for the UAVs at intermediate locations is 2 and 1 s, respectively. The desired minimum separation between the UAVs is $r^d_{\mathrm{v2v}} = 4$ m. As mentioned in Sec. III, there are definitions on minimum separation between unmanned vehicles. However, for the simulation scenarios the values chosen for safe separation between UAVs are specific to those instances and are not standard values. This is for simplicity and could be modified to standard values on implementation. For the mission, the CONCORD system predicts conflicts at the beginning of each time horizon. In this work, we consider the span of time horizon as $t_H = 10$ s. At $t = 1$ s, a conflict is predicted at $t_c = 5.07$ s. That is, time to minimum separation $t_{\mathrm{ms}} = 4.07$ s and $r_{\mathrm{v2v}}(t_c) = 3.928$ m.

Let the vehicles use speed control for achieving avoidance [59]. The resolution actions in the examples are speed variations. This is chosen solely for the purpose of demonstration. There could be other actions for avoidance. For example, if the avoidance is achieved by alternatively applying speed and heading change, the metric for correlated equilibrium could be path length, or resultant control effort, or even time of avoidance.

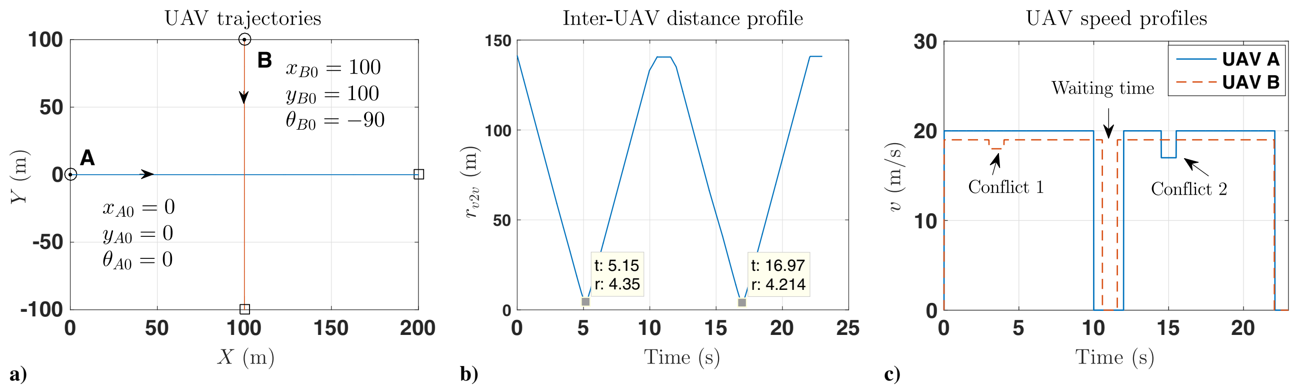


Fig. 13 a) UAV trajectories. b) Inter-UAV distance profile. c) Speed profiles.

The resolutions computed for the above case are (20, 18) m/s and (21, 19) m/s. For the conflict resolution game, these are the actions for the players. Now, consider the payoff/utility function $U^A = 0.5(v_A^2 - v_{A\text{avo}}^2)$ and $U^B = 0.5(v_B^2 - v_{B\text{avo}}^2)$. This is the energy incurred by an agent for avoidance. The utility function captures the energy spent for avoidance by a UAV. Its value is negative when it speeds up, whereas it is positive when it slows down. The objective is to minimize any additional energy spent for avoidance. The payoff matrix is obtained as shown in Table 9. As B slows down, its energy is positive and vice versa. The objective function is chosen to be the social welfare function [60] given by

$$J = \min \sum_{i=1}^{4} p_i (U^A + U^B) \tag{20}$$

This objective function is similar to expression given in Eq. (11), where $b = U^A + U^B$, the social welfare function. The welfare function computes the sum of utilities corresponding to each combination of actions and finds the expected value over all actions in the joint action space. This definition remains the same for all the optimizations performed. The social welfare function is employed as it tries to optimize the total cost of the players without regard to how the payoffs are distributed among the players. The element that changes is the utility or the payoff function of the individual players. Different UAVs having different utility function is a realistic assumption as this would ideally represent priorities assigned to UAV operators.

The correlated equilibrium is found to be the pure strategy, (20, 18) m/s. The CONCORD system sends resolutions to vehicles and they execute the commands to maintain separation. Now, the UAVs proceed to their goal locations. They reach their destinations at the end of 9.99 and 10.57 s, respectively. The return travel puts the vehicles again in conflict. At 13 s, the CONCORD system predicts collision 3.77 s ahead of time. The prediction is sent to the vehicles and UAVs send resolutions to the CONCORD system. The payoffs are computed as shown in Table 10. The correlated equilibrium payoff is (17, 19) m/s. The resulting inter-UAV distance profile is shown in Fig. 13b, where it can be seen that the minimum separation between the vehicles always remains above 4 m. The UAV speed profiles are shown in Fig. 13c, which shows the speed changes during conflicts and the waiting period. The mission is completed at the end of 22 s.

Table 9 Payoff matrix for first trip

A ↓ B →	19	18
20	0, 0	0, 18.5
21	−20.5, 0	−20.5, 18.5

Table 10 Payoff matrix for return trip

A ↓ B →	19	22
20	0, 0	0, −61.5
17	55.5, 0	55.5, −61.5

If we examine the repeated games in this example, conflict 1 is resolved by UAV B decreasing its speed, and conflict 2 by UAV A reducing its speed. It is evident that, if any of the UAVs deviate, collision will occur. Energy expended for the to-and-fro mission for UAV A is

$$E_A = \frac{1}{2}\int_{t=0}^{t=22.09} v_A^2 \,\mathrm{d}t = \frac{1}{2}(20^2 \times 10 + 0^2 \times 2 + 20^2 \times 2.5 + 17^2 \times 1 + 20^2 \times 6.59) = 3962.5 \text{ m}^2/\text{s}$$

Similar computation for UAV B gives

$$E_B = \frac{1}{2}\int_{t=0}^{t=22.05} v_B^2 \,\mathrm{d}t = \frac{1}{2}(19^2 \times 3 + 18^2 \times 1 + 19^2 \times 6.57 + 0^2 \times 1 + 19^2 \times 10.48) = 3619.025 \text{ m}^2/\text{s}$$

The percentage difference in energy expended for avoidance for UAVs is computed by

$$\Delta E^* = \frac{\bar{E}^* - E^*}{\bar{E}^*}$$

where $\bar{E}$ corresponds to energy expenditure for mission had there been no conflict, and asterisk (*) denotes UAV A or B. The ΔE for UAVs A and B are 0.6 and 0.4%, respectively. If the vehicles continue their mission, the average energy expenditure of the vehicles in repeated games would equalize. Thus, the CONCORD system helps early collision detection and ensures vehicle safety in a fair manner.

The avoidance algorithm used to compute resolutions is not the main focus in these examples. Given the resolutions for avoidance using any available avoidance algorithm, how the CONCORD system computes strategies considering the UAV priorities for fair deconfliction is the main theme of the paper. So, details of avoidance algorithm are not included. The computations for avoidance control for one of the examples are now included in the Appendix for completeness.

In this example, the UAVs avoid conflict by UAV B slowing down in the first run. In the return trip, UAV A slows down to avoid conflict. In both cases, one of the UAVs has slowed down to deconflict. In general, slowing down or speeding up would depend on the vehicle states and their relative separation. The UAVs may not always slow down to achieve avoidance.

3. Multiple-UAV Scenario

This case shows how correlated resolutions are computed when more than two vehicles are on collision course. The game cannot be

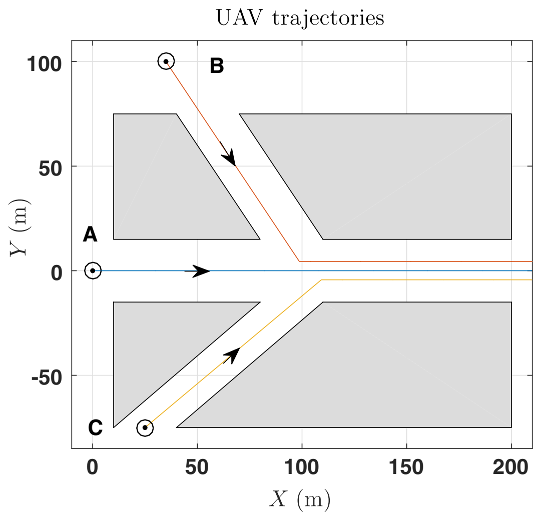


Fig. 14 The multi-UAV conflict scenario.

represented in bimatrix form in this case. Cooperative and non-cooperative scenarios are considered for this multi-UAV scenario.

Consider the multi-UAV scenario as shown in Fig. 14. This example represents a converging traffic, where UAVs A, B, and C, flying at constant speeds, meet at an intersection. The desired minimum separation r^d_{v2v} between UAVs is 10 m.

Case A: UAVs A, B, and C are cooperative. The conflict is predicted at 0.5 s. The detected conflicts are ranked based on the time to minimum separation, t_{ms}. Table 11 shows the vehicle initial conditions, speeds, objectives, conflict time, separation at conflict time, and time to minimum separation associated with them. The conflicts occur simultaneously, about 8 s from the instant of prediction. The payoff/utility functions for the UAVs are inter-UAV distances and mission time, and they are given by

$$U^A = U^C = \sum r^{\min}_{\mathrm{v2v}}(t) \tag{21}$$

$$U^B = t_M \tag{22}$$

Now, the objective of the linear program to compute correlated strategies is the welfare function, given by

$$J = \min \sum_k^w p_k(c_1 1/U^A + c_2 U^B + c_3 1/U^C) \tag{23}$$

where $w = 8$ and c_1, c_2, c_3 are coefficients with appropriate units. We assume the values of these to be equal to 0.33, giving equal weightage to all UAV utilities. The resolutions computed are (6.5, 16.5, 10.5) m/s, (6.5, 16.5, 10.5) m/s, and (7, 17.5, 11) m/s.

Here, A and B are assumed to have computed the same resolution, which could happen in general. CONCORD system uses the control inputs to compute the correlated equilibrium. The game can no longer be represented in bimatrix form. From the resolutions, each UAV has two strategy profiles, represented by S_1 and S_2, given in Table 12. The corresponding payoffs for combinations of these actions are given in Table 12, where the subscripts of U represent the action combination (U_{121} to be read as UAVs A, B, and C choosing 6.5, 17.5, and 10.5 m/s, respectively). Infeasible combinations are given 0 payoff.

The correlated equilibrium is computed as (6.5, 17.5, 11) m/s. This recommendation is sent to the UAVs. They execute the commanded speed changes to remain safe at the intersection. The speed profiles and inter-UAV distance with time are given in Figs. 15a and 15b. The distance profiles show that the vehicles remain safe from collision.

Case B: In this case, let UAV C be non-cooperative. UAV C can be thought of as a high-priority vehicle, which cannot change its flight parameters following the recommendations from CONCORD system but continues to send state updates. The resolution is to be computed for the anticipated action of the non-cooperative UAV. The conflict parameters are same as in Table 11. Consider that UAVs A and B send the same resolutions as before, as they are unaware of the intentions of UAV C. The CONCORD system recomputes the resolutions assuming no speed change for UAV C. The strategies S_1 and S_2 and payoffs are shown in Table 13. The objective function to be minimized is

$$J = \min \sum_k^w p_k(c_1 1/U^A + c_2 U^B) \tag{24}$$

where c_1 and c_2 equal 0.5, and $w = 4$. The correlated equilibrium is found to be (6.5, 17.5, 0) m/s. The corresponding velocity profiles and inter-UAV distance profiles are shown in Fig. 16.

It is interesting to look into the case where two vehicles are non-cooperative instead of one. Other than manual control take-over, an avoidance resolution is difficult in this scenario if the non-cooperative UAVs do not respond to the CONCORD system. If the vehicles respond but their priorities are unknown, the CONCORD system could still compute resolutions by assuming same priority for all the vehicles in conflict.

4. *Multiple UAVs in Structured Traffic Scenario*

This is a typical realistic scenario where UAVs are integrated to the airspace in a structured manner. Such an environment can be envisaged as one with multiple operators deploying UAVs with their own priorities. CONCORD resolves conflicts in this urban environment where the mission profiles of the UAVs are assumed to be known.

Consider four UAVs traveling to their goal locations in an urban environment, as shown in Fig. 17, where the shaded regions correspond to static obstacles. UAVs A, B, C, and D have the mission details and specifications, as given in Table 14. The mission is planned for UAVs A, B, and C such that their paths are nonconflicting. UAV A moves between A_0 and A_1, performing surveillance. UAV B is a multidelivery vehicle that travels to B_3 via B_1 and B_2, and

Table 11 UAV initial conditions and conflict parameters

UAV	(x_0, y_0, θ_0)	Speed, m/s	Objective	t_c, s	UAVs	$r_{\mathrm{v2v}}(t_c)$, m	t_{ms}, s
A	(0, 0, 0)	11	Max–min inter-UAV distance	8.66	AB	9.932	8.16
B	(35, 100, −56.31)	12.5	Min mission time	8.56	AC	9.967	8.06
C	(25, −75, 42)	11.5	Max–min inter-UAV distance	9.2	BC	9.924	8.7

Table 12 Actions and payoffs for the possible combination of actions

S_1	S_2	Payoff	U_{111}	U_{112}	U_{121}	U_{122}	U_{211}	U_{212}	U_{221}	U_{222}
6.5 16.5 10.5	7 17.5 11	**Value**	(24.86, 17.41, 20.73)	(0, 0, 100)	(25.79, 17.26, 21.43)	(26.64, 17.26, 21.66)	(23.01, 17.41, 19.83)	(0, 0, 0)	(0, 0, 0)	(24.8, 17.26, 20.73)

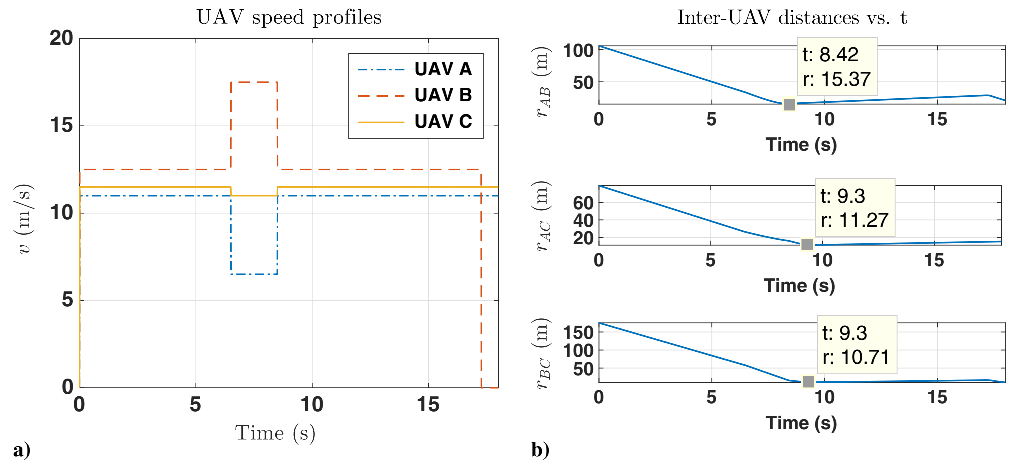


Fig. 15 a) UAV speed profile. b) Inter-UAV distances with time, for the cooperative multi-UAV case.

Table 13 Actions and payoffs for the possible combination of actions for non-cooperative UAV C

S_1	S_2	Payoff	U_{11}	U_{12}	U_{21}	U_{22}
6.5 17.5	7 18	Value	(27.54, 17.26)	(27.87, 17.19)	(25.70, 17.26)	(25.05, 17.19)

performs a delivery action there, followed by deliveries at B_2 and B_1, respectively, on its return journey. This is simulated by a waiting time of 2 s at each of these locations. UAV C does a one shot travel from C_0 to C_1. A non-cooperative or a rogue UAV D resumes its mission at $t = 5$, its objective being unknown to the ground station and other UAVs.

The ground control station receives the state update and predicts conflicts at the beginning of each time horizon. UAVs A and B resume their mission simultaneously rather than the preplanned staggered start. This results in conflicts between them. The specific parameters of the conflicts in each time horizon are listed in Table 15.

The payoffs for the vehicles are computed based on the individual vehicle objectives, mentioned in Table 14. Within the first prediction horizon, two conflicts are predicted, out of which the one between UAVs B and C is prioritized due to lesser t_{ms}. The UAVs compute their resolution based on their respective priorities and are sent to the CONCORD system. The UAVs follow speed modulations for avoidance and the normal form of the game, based on their resolutions shown in Table 16. The first entry of each cell is the inter-UAV distance achieved and second is that of the mission time. UAV B maximizes $r_{\mathrm{v2v}}^{\min}$ and UAV C minimizes t_M, while avoiding each other.

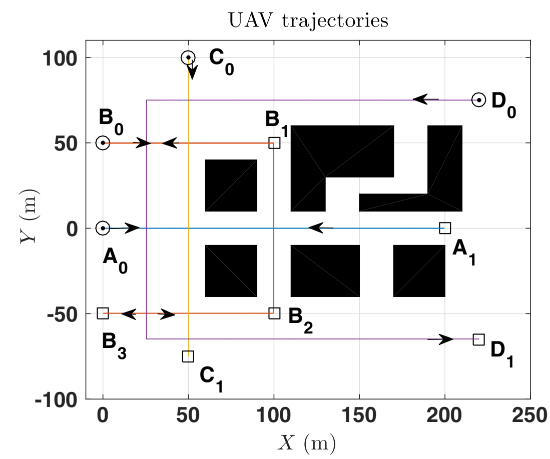


Fig. 17 The multi-UAV structured traffic scenario in a typical urban environment.

The infeasible solution is assigned zero. Formulating the linear program to minimize the welfare function, the equilibrium is found to be a pure strategy, which is (13.5, 7.5) m/s. UAVs execute the recommendation starting from $t = 1$ to $t = 3$ s.

The normal form of the game between A and B, for the next conflict resolution, is given by Table 17. The conflict prediction by the CONCORD system proves to be helpful in scenarios like that between UAVs A and B, which is difficult to be detected by on-board sensors in advance due to obstacles in their vicinity. UAV A

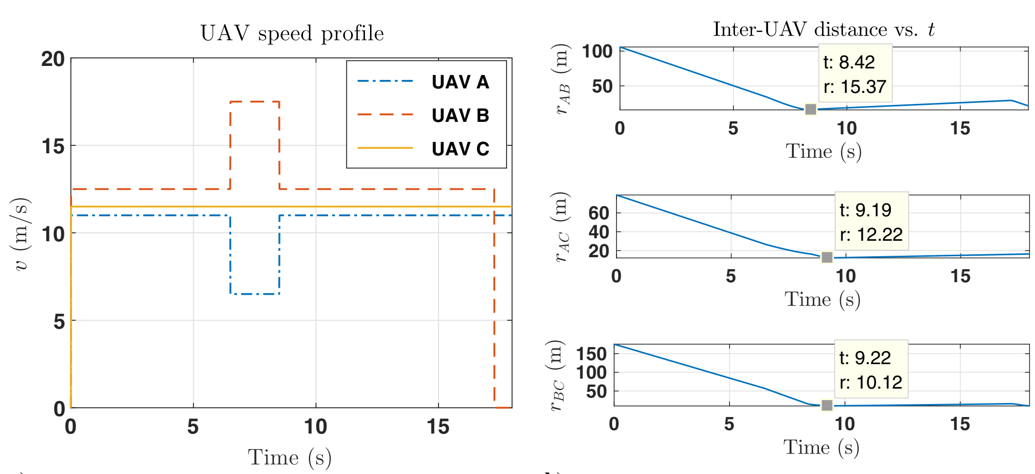


Fig. 16 Non-cooperative multi-UAV scenario: a) UAV speeds; b) inter-UAV distances with time.

Table 14 UAV mission details and specifications in multi-UAV structured traffic scenario

UAV	(x_0, y_0, θ_0)	Objective	Mission type	Path	r_v^d, m	Speed, m/s
A	(0, 0, 0)	Min energy	Surveillance	$A_0 - A_1$-repeat	5	10.5
B	(0, 50, 0)	Max–min inter-UAV separation	Multidelivery	$B_0 - B_1 - B_2 - B_3$-return	3	15
C	(50, 100, $3\pi/2$)	Min mission duration	Delivery	$C_0 - C_1$	4	16
D	(220, 75, π)	Unknown (non-cooperative)	Unknown	$D_0 - D_1$	5	17

Table 15 Predicted conflicts in different time horizon

		Conflict in t = 0–10 s				Conflict in t = 20–30 s			
UAVs	r_{v2v}^d, m	t_c	$r_{v2v}(t_c)$	t_{ms}	Rank	t_c	$r_{v2v}(t_c)$	t_{ms}	Rank
AB	8	9.66	7.916	9.16	2	——	——	——	——
AC	9	——	——	——	——	——	——	——	——
AD	10	——	——	——	——	——	——	——	——
BC	7	2.93	6.807	2.43	1	——	——	——	——
BD	8	——	——	——	——	23.57	7.881	3.07	1
CD	9	——	——	——	——	——	——	——	——

Table 16 Payoff matrix for UAVs B and C

B ↓ C →	**7.5**	**18**
13.5	7.17, 11.98	7.21, 10.66
16	10.83, 11.98	0, 0

Table 17 Payoff matrix for UAVs A and B

A ↓ B →	**13**	**15**
10.5	0, 8.21	100, 100
12	−16.875, 10.67	−16.875, 8.38

minimizes the energy expenditure for avoidance. Computing the equilibrium, the correlated strategy is found as (10.5, 13) m/s. The vehicles execute the command between $t = 7$ s and $t = 9$ s. Next, conflict is predicted in the third prediction horizon between B and D. UAV D, being a non-cooperative player, is assumed to send state updates to the CONCORD system but not follow its recommendations. For this scenario, UAV B computes the resolution as shown in Table 18. The infeasible solution is assigned zero payoff. UAV B slows down to 10 m/s for a duration of 1.62 s. Any speed below 10 m/s is a safety policy for UAV B, for speeds of UAV D equal to 17 m/s or more. The CONCORD system computes the correlated equilibrium strategy and recommends 10 m/s, which ensures safe movement of the UAVs. The UAV speeds for the duration of the mission are shown in Fig. 18a. The UAV speed changes during the avoidance could be observed on this profile. The speed of UAV B falls to zero upon reaching the delivery points. The inter-UAV distances between the UAV pairs are shown in Fig. 18b. It can be seen that the UAVs maintain the desired separation between each other and complete their travel. An animation showing the evolution of UAV trajectories can be found in the link https://youtu.be/Rfy1hNWuwzw. The CONCORD system helps to avoid immediate conflicts with early predictions. The low computational cost of the decision-making mechanism helps in quick resolutions. Also, the CONCORD system, being an intermediate decision maker, facilitates collision avoidance in a fair manner, considering the priorities of all the vehicles involved.

Table 18 Payoff matrix for UAVs B and D

B ↓ D →	**17**
10	8.78, 100
15	100, 100

Monte Carlo simulations: The urban scenario considered above is randomized to examine the effectiveness of the CONCORD algorithm. The UAVs are located at the same points but their speeds follow normal distribution with mean value equal to the nominal speed considered above and deviation of three sigma. The speeds of UAVs A and B, A and C, and B and C are varied for this study. UAV D is a non-cooperative UAV and its speed is assumed to remain unchanged. A total of 100 runs were carried out for each of the UAV pairs. The results of the simulations are as following.

Set 1: Figure 19 shows the simulation results for random speed values assigned to UAVs A and B. The number of conflicts occurring in each run of the simulation is shown in Fig. 19a. It can be seen that the maximum number of conflicts is 3. There were 7 cases in which the UAVs were not in conflict. Figure 19b shows the speed profiles of UAVs A and B used in the simulation. Two instances of data from the simulation are given in Table 19. In iteration 3, UAVs B and A get in conflict twice. For both of these, deconfliction is achieved such that no vehicle has to speed up on all the conflict occasions. The energy expended, minimum inter-UAV separation, and mission duration for UAVs A, B, and C are optimized while achieving this resolution. UAV C maintains its speed in both the instances of iteration 74 as it is not out of conflict with UAV B when conflict with UAV A is detected. So, UAV A adjusts speed to maintains safe separation in this case.

Set 2: Figure 20 shows the simulation results where speeds of UAVs A and C are randomized. The number of conflicts in each run of the simulation is shown in Fig. 20a. The maximum number of conflicts that occurred is 4. In all the 100 runs, there were conflict instances. Figure 20b shows the speed profiles of UAVs A and C used in the simulation. Similar observations on fair deconfliction, as in Set 1, can be observed here (Table 20). UAV B has three resolutions in iteration 13, where its speed is increased and decreased alternatively. Similarly, other UAVs also avoid conflict.

Set 3: In this set, UAVs A, B, and C are assigned random speed values for 125 runs. One set of data is given in Table 21 for this set of simulations. This is a particular case in which UAV A continuously remains on collision course with other UAVs. For this iteration, the speeds of UAVs B and C are comparable. The close proximity of UAV paths and comparable speeds resulted in this conflict scenario. Nevertheless, CONCORD takes care of the deconfliction in a fair manner.

5. *Comparison with Other Methods*

Monte Carlo simulations are performed using three other methods to compare the performance of the proposed CONCORD algorithm. The same urban scenario is considered as the initial condition. The simulation results are given below.

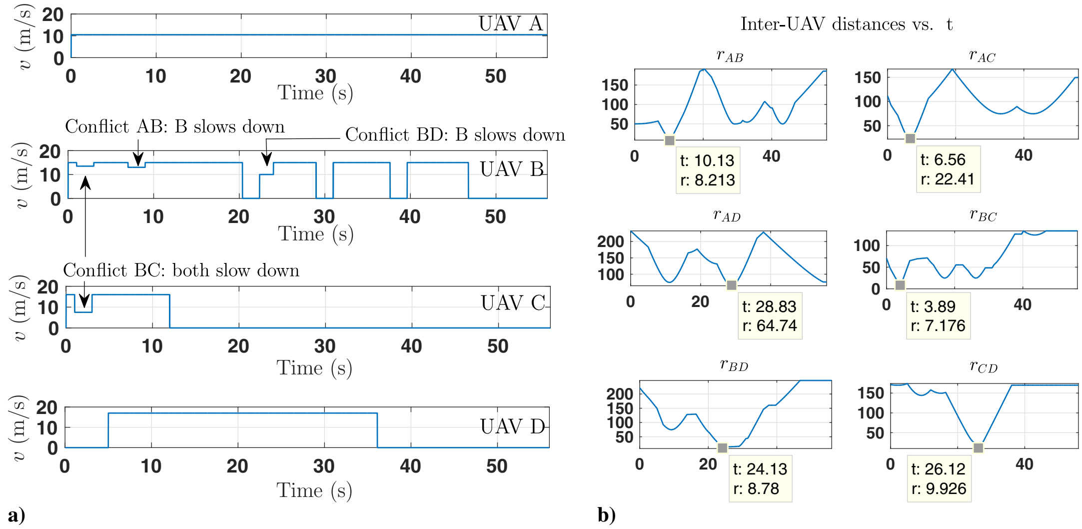


Fig. 18 a) UAV speeds. b) Inter-UAV distance profile for structured multi-UAV case.

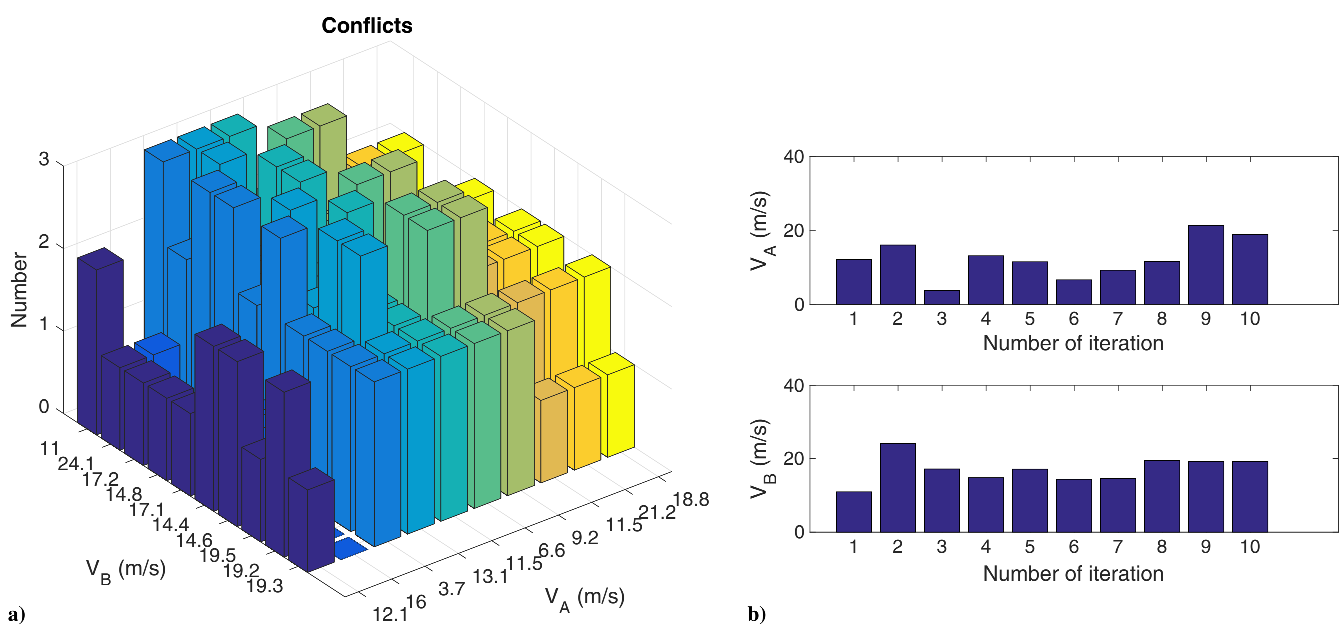


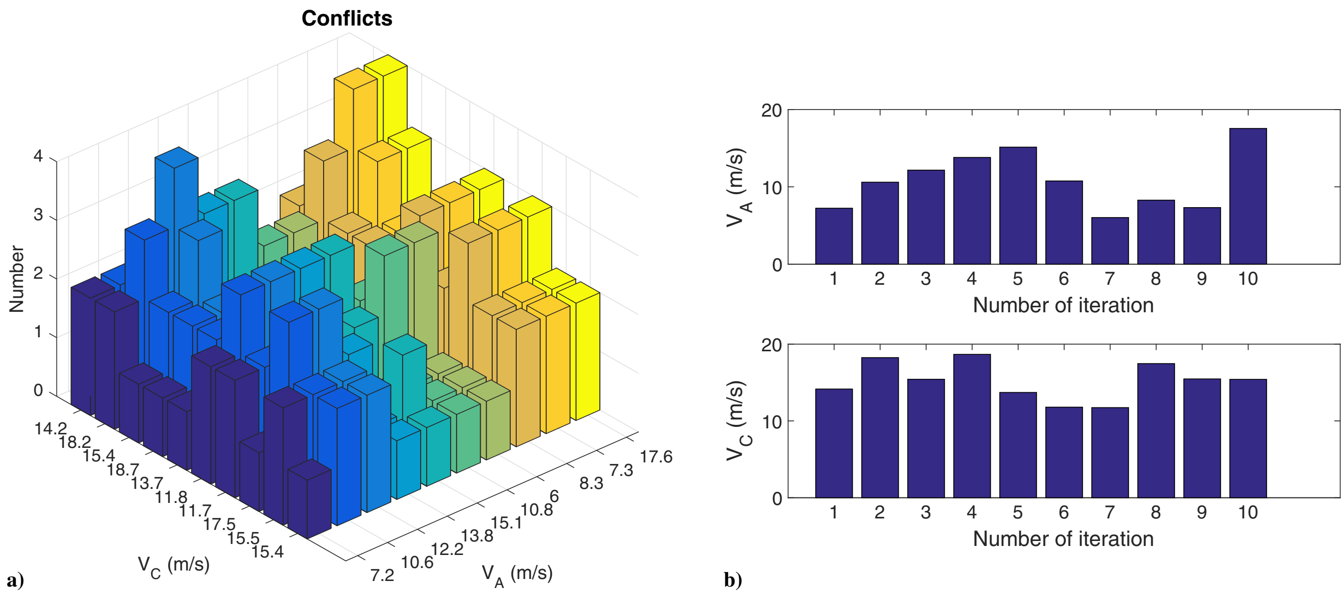
Fig. 19 Results of Monte Carlo simulations for Set 1: a) number of conflicts occurred in each run; b) speed profile used for the simulations.

Table 19 Data from Monte Carlo simulations for Set 1

Iteration no.	No. of conflicts	Conflict between	UAV nominal speed, m/s	UAV avoidance speeds, m/s
3	3	BC	(12.11, 17.17, 16, 17)	$(v_B, v_C) = (19.67, 12.00)$
		AB		$(v_A, v_B) = (16.11, 12.67)$
		BD		$(v_B, v_D) = (11.67, 17.00)$
74	3	BC	(11.52, 14.81, 16, 17)	$(v_B, v_C) = (12.81, 21)$
		AC		$(v_A, v_C) = (13.02, 21)$
		BD		$(v_B, v_D) = (12.81, 17)$

Random avoidance strategy: First, conflicts are resolved using random actions taken by the UAVs. Here, the UAVs chose any random speed to avoid a conflict. The speed profiles used are the same as in Set 1 and Set 2, respectively. The results obtained are given below. Figures 21a and 21b show the conflicts that occurred and those that were resolved when UAVs A and B randomize their speeds. It is observed that the number of conflicts that occurred during the 100 runs is more but not all conflicts are resolved. This is due to the randomness in chosen avoidance actions. A similar trend can be observed for UAVs A and C (Fig. 22). The maximum number of conflicts that have occurred is 8 and 10, respectively. Data from an iteration for each of the Monte Carlo simulations are given in Tables 22 and 23, respectively. It can be seen that the speed variations have no considerations over the utility functions of the UAVs. Also, in contrast to the observations in Set 1 simulations, when UAVs end up in conflict multiple times, same vehicle is observed to expend energy to get out of conflict. This makes the conflict resolution unfair.

Speed control strategy: Now, conflicts are resolved using the speed control algorithm [59]. No correlated equilibria are computed, and the speed variations are directly recommended to the vehicles. The speed profiles used are the same as in Set 1 and Set 2, respectively. Figure 23a shows the number of conflict that occurred during 100 runs for UAVs A and B randomizing their speeds. The maximum number of conflicts was found to be 5. Similar to Set 1, seven iterations resulted in no-conflict situation. Data corresponding to iteration 3 are given in Table 24. The speed recommended for UAV B is different from that of the CONCORD resolution in Set 1. It can be



a) b)

Fig. 20 Results of Monte Carlo simulations for Set 2: a) number of conflicts occurred in each run; b) speed profile.

Table 20 Data from Monte Carlo simulations for Set 2

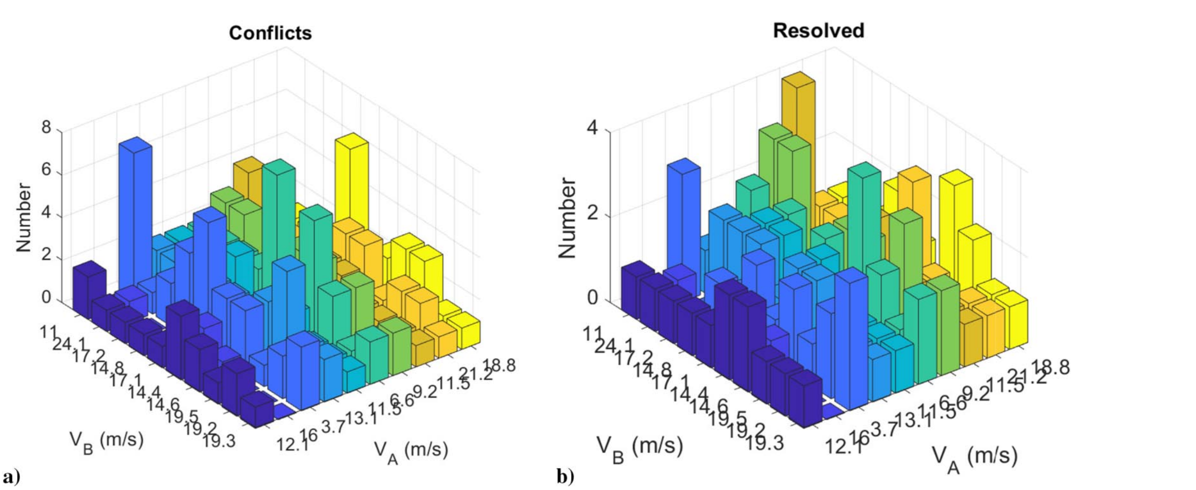
Iteration no.	No. of conflicts	Conflict between	UAV nominal speed, m/s	UAV avoidance speeds, m/s
13	4	BC	(10.59, 15, 15.42, 17)	$(v_B, v_C) = (12.50, 21.42)$
		AB		$(v_A, v_B) = (5.59, 21.42)$
		BD		$(v_B, v_D) = (17.50, 17.00)$
		AB		$(v_A, v_B) = (9.59, 16.00)$
56	2	BC	(10.75, 15, 11.79, 17)	$(v_B, v_C) = (11.75, 13.5)$
		AC		$(v_A, v_C) = (16.50, 17.0)$

Table 21 Data from Monte Carlo simulations for Set 3

Iteration no.	No. of conflicts	Conflict between	UAV nominal speeds, m/s	UAV speeds, m/s
35	4	BC	(10.25, 15.30, 15.21, 17.0)	$(v_B, v_C) = (12.3, 22.41)$
		AC		$(v_A, v_C) = (7.25, 22.41)$
		AB		$(v_A, v_B) = (7.25, 17.31)$
		AC		$(v_A, v_C) = (7.25, 13.41)$

observed that UAV B gets on a collision course thrice and its resolved by alternatively speeding up and slowing down. However, if the utility function for UAV B is evaluated (Table 25), it is found to have an optimal value with CONCORD than with speed changing strategy. The mission duration of UAV C in each of these cases is 11.27 and 11.66 s, respectively. The first conflict game of iteration 3 is resolved in a fairer manner by the CONCORD, whereas it is solved in favor of UAV C in the speed-changing strategy. In other words, UAV B could have achieved a better payoff than the one obtained with speed-changing strategy for approximately the same payoff achieved by UAV C. The same happens in the second game also. This demonstrates the effectiveness of the CONCORD framework.

Similarly, randomizing the speeds of UAVs A and C, with the speed distribution same as that in Set 2, is considered next. The maximum number of conflicts that occurred is 4. All iterations resulted in conflicts. If we consider iteration 13 as in Set 2, four conflicts are found to occur in this case as well. The resolutions corresponding to it are shown in Table 26. UAV B gets in conflict four times. Similar to the previous observation, Table 27 suggests that the utility is better optimized for the UAVs while using CONCORD. Mission duration for UAV C is 10.75 s without CONCORD and 9.405 s with CONCORD. Energy expended by UAV A is 1043 m^2/s, whereas with CONCORD, it is 887.6 m^2/s. Hence, the CONCORD



a) b)

Fig. 21 Results of Monte Carlo simulations with UAVs A and B using random speeds to avoid: a) number of conflicts occurred in each run; b) number of conflicts resolved.

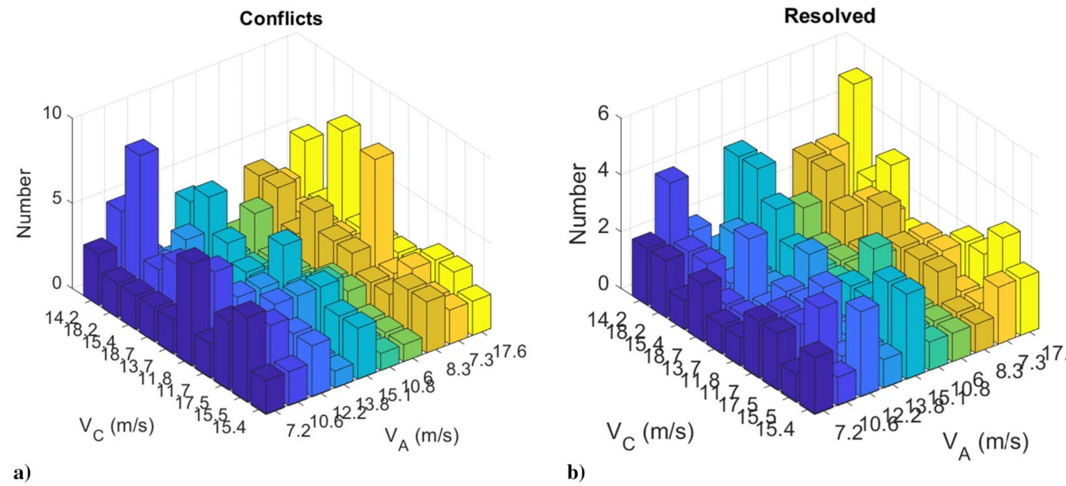


Fig. 22 Results of Monte Carlo simulations with UAVs A and C using random speeds to avoid: a) number of conflicts occurred in each run; b) number of conflicts resolved.

Table 22 Data from Monte Carlo simulations for UAVs A and B with random resolutions

Iteration no.	No. of conflicts	Conflict between	UAV nominal speed, m/s	UAV avoidance speeds, m/s
35	4	BC	(14.8, 3.7, 16, 17)	$(v_B, v_C) = (15.6562, 15.3414)$
		AB		$(v_A, v_B) = (5.1214, 14.1410)$
		AD		$(v_A, v_D) = (1.56, 17.00)$
		AB		$(v_A, v_B) = (8.463, 5.142)$

Table 23 Data from Monte Carlo simulations for UAVs A and C with random resolutions

Iteration no.	No. of conflicts	Conflict between	UAV nominal speed, m/s	UAV avoidance speeds, m/s
22	3	AB	(11,15,11.5, 17)	$(v_A, v_B) = (8.8247, 22.9216)$
		AC		$(v_A, v_C) = (20.8247, 10.0134)$
		BC		$(v_B, v_C) = (22.9216, 9.0134)$

framework is found to give fair resolutions and better payoff than the speed-control-based deconfliction strategy.

Nash-equilibrium-based speed control strategy: A third method is also simulated for comparison, where the UAVs resolve conflict by computing Nash equilibria speeds. The number of conflicts for 100 runs of Monte Carlo simulations for UAVs A and B and that for UAVs A and C randomizing their speeds are shown in Fig. 24.

Figure 24a shows the case where UAVs A and B randomize their speeds. The maximum number of conflicts observed is 3. The data corresponding to iteration 3 are given in Table 28. It is found that Nash equilibrium solution matches with the CONCORD solution in this case. In this case, both the methods are found to perform equally well. For iteration 74, CONCORD is found to have a better performance than the Nash-equilibrium-based strategy. Table 25 shows that CONCORD has a slight improvement over Nash-equilibrium-based solution. UAV C also has a better utility with CONCORD than Nash equilibrium strategy in this case.

Figure 24b shows the case where UAVs A and B randomize their speeds. The maximum number of conflicts observed is 4. Data for iteration UAVs A and C are given in Table 29. UAV A gets on collision course twice, and UAV B ends up in conflict four times. Comparing the resolutions computed using Nash equilibrium and that of correlated equilibrium, it is evident that UAV B had to speed up consecutively. UAV B achieves better payoff with CONCORD than with Nash equilibrium (Table 27) although Nash-equilibrium-based

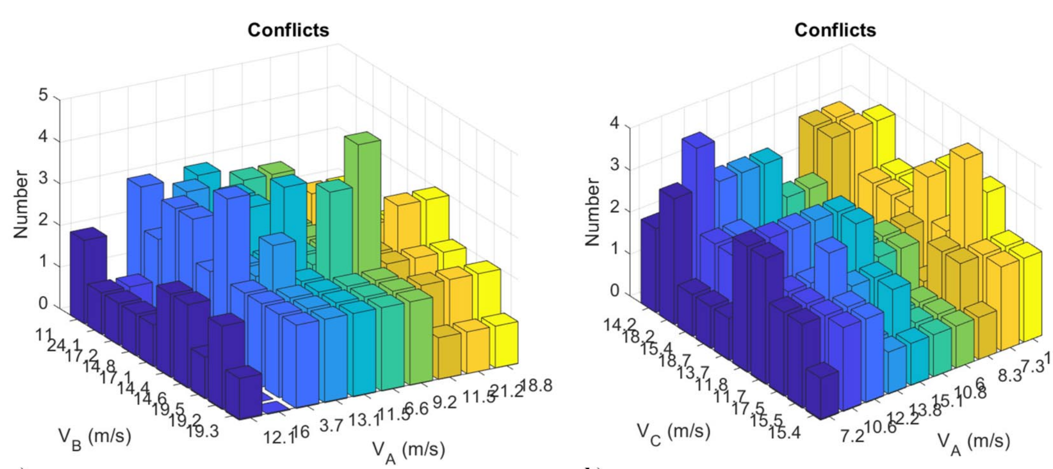


Fig. 23 Results of Monte Carlo simulations with only speed control for a) UAVs A and B, and b) UAVs A and C.

Table 24 Data from Monte Carlo simulations for UAVs A and B where speed control is used for deconfliction

Iteration no.	No. of conflicts	Conflict between	UAV nominal speed, m/s	UAV avoidance speeds, m/s
3	3	BC	(12.11, 17.17, 16 17)	$(v_B, v_C) = (19.67, 14)$
		AB		$(v_A, v_B) = (16.11, 13.17)$
		BD		$(v_B, v_D) = (10.17, 17.00)$

Table 25 Payoff achieved by UAV B using speed control strategy and the CONCORD (UAVs A and B randomizing)

UAVs	Max–min r_{v2v} with speed control strategy, m	Max–min r_{v2v} with CONCORD, m	Max–min r_{v2v} with Nash equilibrium strategy, m
AB	15.62	16.35	13.66
BC	11.66	16.33	11.65
BD	8.40	7.94	9.77

With CONCORD the payoff is better and fairer among UAVs.

Table 26 Data from Monte Carlo simulations for UAVs A and C where speed control is used for deconfliction

Iteration no.	No. of conflicts	Conflict between	UAV nominal speed, m/s	UAV avoidance speeds, m/s
13	4	BC	(10.59, 15, 15.42, 17)	$(v_B, v_C) = (12.50, 18.42)$
		AB		$(v_A, v_B) = (11.59, 16.92)$
		BD		$(v_B, v_D) = (17.50, 17.00)$
		AB		$(v_A, v_B) = (11.59, 17.16)$

Table 27 Payoff achieved by UAV B (UAVs A and C randomizing) with speed control strategy and with CONCORD

UAVs	Max–min r_{v2v} with speed control strategy, m	Max–min r_{v2v} with CONCORD, m	Max–min r_{v2v} with Nash equilibrium strategy, m
AB	13.42	9.12	11.14
BC	10.58	17.93	12.73
BD	8.07	9.35	8.12

Table 28 Data from Monte Carlo simulations for UAVs A and B where Nash equilibrium is used to compute resolutions

Iteration no.	No. of conflicts	Conflict between	UAV nominal speed, m/s	UAV avoidance speeds, m/s
3	4	BC	(12.11, 17.17, 16, 17)	$(v_B, v_C) = (19.67, 14.00)$
		AB		$(v_A, v_B) = (16.11, 13.17)$
		BD		$(v_B, v_D) = (10.17, 17.00)$
74	3	BC	(11.52, 14.81, 16, 17)	$(v_B, v_C) = (13.17, 18)$
		AC		$(v_A, v_C) = (15.13, 18)$
		BD		$(v_B, v_D) = (12.81, 17)$

Table 29 Data from Monte Carlo simulations for UAVs A and C, where Nash equilibrium is used to compute resolutions

Iteration no.	No. of conflicts	Conflict between	UAV nominal speed, m/s	UAV avoidance speeds, m/s
13	4	BC	(10.59, 15, 15.42, 17)	$(v_B, v_C) = (12.50, 18.42)$
		AB		$(v_A, v_B) = (13.29, 18.36)$
		BD		$(v_B, v_D) = (21.50, 17.00)$
		AB		$(v_A, v_B) = (8.43, 22.68)$

solution outperforms the speed-based avoidance strategy. The mission duration for UAV C is 10.35 s, which is more than 9.41 s, achieved via CONCORD.

In summary, the CONCORD framework is found to carry out deconfliction such that all involving vehicles achieve safety while optimizing their priorities. It definitely outperforms the random strategy in terms of number of deconflictions and UAV payoff. CONCORD has better performance than the speed-based strategy in achieving optimal utilities for the players while carrying out avoidance. CONCORD improves on Nash equilibrium solutions as well, even though their performances may be comparable at times.

VI. Conclusions

This work presented the CONCORD system, a semi-autonomous decision-making system for conflict resolution. The proposed framework is based on the game-theoretic concept of correlated equilibrium. Correlated equilibrium is a generalized concept that is computationally tractable for any number of players. The UAVs are considered players, and a collision avoidance game is formulated. The payoff matrices are computed based on the priorities of vehicles. The presented architecture can handle cooperative, non-cooperative, and multiple UAVs. The

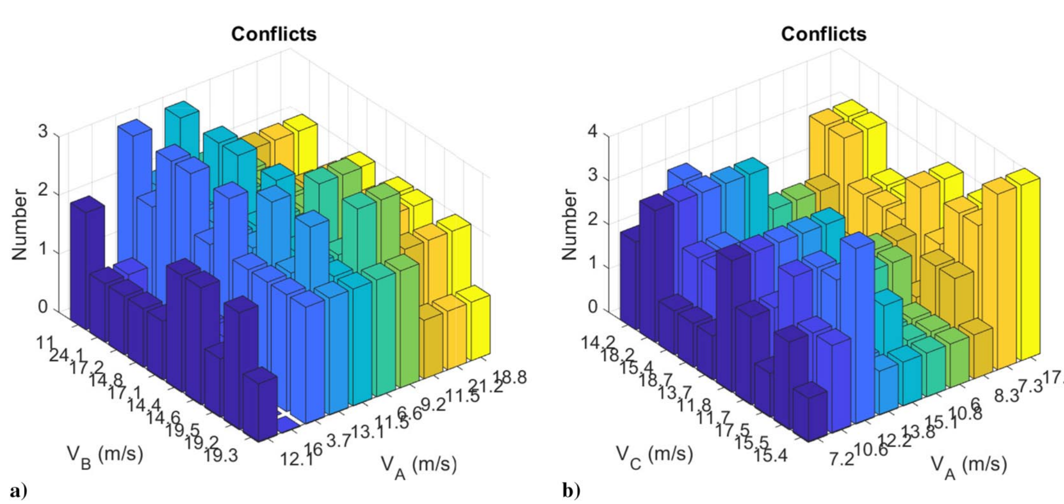


Fig. 24 Results of Monte Carlo simulations with Nash-equilibrium-based resolution for a) UAVs A and B and b) UAVs A and C.

realistic simulation test cases show the advantages of having a mediated system resolve collision, which otherwise becomes a complex procedure. The CONCORD system could be used as a conflict resolution module for any UAV traffic management system and any avoidance algorithm that the vehicles use. Further, integrating the CONCORD system to UAV traffic management systems is also discussed, which describes how the CONCORD system communicates with the central traffic management authority and service providers to execute its mission. Comparison with two baselines and Nash-equilibrium-based strategy demonstrated the better performance of the proposed avoidance framework.

Few future extensions to this work are the following. For vehicles that do not follow the recommendations, there needs to be a mechanism to ensure safety, and such a mechanism is currently absent. Also, the proposed architecture is a semi-autonomous one. However, full automation of the framework is necessary to handle high-density air traffic. The proposed architecture is a centralized approach that relies on UAV-ground station communication, which could be noisy or delayed. A method needs to be devised to handle such system-level disturbances to ensure safety. Verifying the functioning of the proposed algorithm using real experimental data is an immediate future extension. Field experiments involving CONCORD-based deconfliction would be the subsequent step in proving effectiveness of the proposed framework.

Appendix: Collision Avoidance Using Speed Control

The collision avoidance algorithm employed for the simulations uses speed modulation as the avoidance control input. Consider the case of two UAVs (Fig. A1) in repeated conflict (Sec. V.2). When conflict is predicted, the following calculations are made (refer [59] for more details). For UAVs A and B, following are the initial conditions: $x_0 = 100$, $y_0 = 100$, $V_A = 20$ m/s, $\theta_A = 0°$, $V_B = 19$ m/s, and $\theta_A = -90°$. The following parameters are computed using the initial conditions:

$$Q = \frac{V_B \cos\theta_B - V_A \cos\theta_A}{V_B \sin\theta_B - V_A \sin\theta_A} \tag{A1}$$

$$\begin{aligned} A &= \frac{2x_0 y_0 (Q^2 - 1) + 2Q(y_0^2 - x_0^2)}{(Q^2+1)^2} \\ &= \frac{19 \times \cos(-90°) - 20 \times \cos 0°}{19 \times \sin(-90°) - 20 \times \sin 0°} \\ &= 1.0526 \end{aligned} \tag{A2}$$

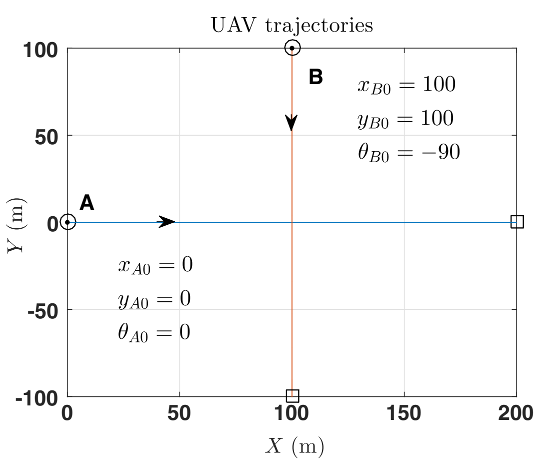


Fig. A1 Two UAVs in conflict scenario.

$$\begin{aligned} B &= \frac{V_B \cos\theta_B \sin\theta_A - V_B \sin\theta_B \cos\theta_A}{(V_B \sin\theta_B - V_A \sin\theta_A)^2} \\ &= \frac{19\cos(-90°)\sin 0° - 19\sin(-90°)\cos 0°}{(19\sin(-90°) - 20\sin 0°)^2} \\ &= 0.0526 \end{aligned} \tag{A3}$$

$$\begin{aligned} C &= \frac{V_A \cos\theta_A \sin\theta_B - V_A \sin\theta_A \cos\theta_B}{(V_B \sin\theta_B - V_A \sin\theta_A)^2} \\ &= \frac{20\cos 0° \sin(-90°) - 20\sin 0° \cos(-90°)}{(19\sin(-90°) - 20\sin 0°)^2} \\ &= -0.055 \end{aligned} \tag{A4}$$

To compute desired speed modulations for collision avoidance between UAVs, we have the following relations:

$$\frac{\partial r_{\min}^2}{\partial V_A} = A \times B; \qquad \frac{\partial r_{\min}^2}{\partial V_B} = A \times C \tag{A5}$$

where $r_{\min}$ is minimum inter-UAV separation. Examining the signs of A and B, UAV A should increase its speed to increase minimum inter-UAV separation. Similarly, examining signs of A and C, UAV B should reduce its speed to increase inter-UAV separation. Thus, the options for UAV A and B are chosen accordingly to compute utility values and hence correlated equilibrium resolution. A similar method is followed for the other simulations as well.

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